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Complex Systems Modelling of Alcohol-related Health Inequalities



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List of Acronyms

ABM Agent-based Modelling 1

CS Complex System 7

CLI Command Line Interface 24

AHP Alcohol Harm Paradox 1

AUD Alcohol Use Disorder 4

DQ Deprivation Quintile 2

FCT Fundamental Cause Theory 1

LHS Latin Hypercube Sampling 33

RHPC Repast for High Performance Computing 24

R4P Repast for Python 6

SES Socioeconomic Status 1

MBSSM Mechanism-Based Social Systems Modelling 8

MPI Message Passing Interface 24

UML Unified Modelling Language 6

WHO World Health Organisation 2

IMD Index of Multiple Deprivation 2

Nomenclature

$\vec{x}$ Agent State Vector: The state vector attributed to each agent.

$\vec{u}_m$ Agent Input Vector: The input vector attributed to each agent.

δ State Transition Function: The state transition function attributed to each agent, mapping inputs to outputs.

$\mathcal{C}$ Communicator: Denotes the communicator.

$\mathcal{A}$ Set of Agents: The set of all agents in the simulation.

$\mathcal{A}_\mathcal{C}$ Set of Communicated Agents: The set of agents which receive communicated information.

N Number of Agents: The number of agents in each simulation, equivalent to $|\mathcal{A}|$.

$I(\cdot)$ Information Variable: A variable which is a fixed-length binary string of information.

$G(\mathbb{V}, \mathbb{E})$ The Graph: The Graph G with vertices $\mathbb{V}$ and edges $\mathbb{E}$.

k Time Steps: The discrete-time variable.

$\mathbb{C}$ Communicated Information: The set of information which is communicated to an agent.

$\mathbb{U}$ Unsolved Information: The set of information where the agent attempted to solve the information and failed.

$\mathbb{S}$ Solved Information: The set of information which the agent attempted to solve and succeeded.

$\mathcal{S}$ Swap Set: The set of agents that are due to swap up $\mathcal{S}_u$ or down $\mathcal{S}_d$.

$P_s(\cdot)$ Swap Probability: Used to denote the probability of an agent swapping up or down.

$\rho(c)$ Agent Risk: The risk an individual agent experiences due to the consumption c of alcohol.

$\bar{\rho}(c, x_1)$ Modified Agent Risk: An equation that takes the original risk function $\rho(c)$ and modifies it to consider age x_1, solved events $|\mathbb{S}|$, unsolved events $|\mathbb{U}|$ and total communicated events $|\mathbb{C}|$

Φ Set of Parameters: The set of experimental parameters used to explore the model.

γ_i Experiment Gradient: A variable corresponding to the resulting gradient of the ith experiment.

$\vec{\tau}_i$ Experiment Counts: Counts the outputs of the related experiments, containing information about every Deprivation Quintile.

Abstract

The Alcohol Harm Paradox is a phenomenon in which those with a low socioeconomic status consistently experience more alcohol-related harms across their lifetime than those with high socioeconomic status, even when drinking the same or less alcohol. It has been proposed that Fundamental Cause Theory, a theory that purposes that socioeconomic factors such as education, wealth, social power, and social prestige expose individuals to disease and can be thought of as the fundamental cause of disease. In this dissertation, a previously developed Agent-based Model is replicated using a software architecture referred to as Mechanism-based Social Systems Modelling, which facilitates the implementation of social theories, such as Fundamental Cause Theory, into an Agent-based Modelling context. 200 total simulations were run in total, where the parameter space was explored using a Latin Hypercube Sample. The Alcohol Harm Paradox was found in the model, but only once in all 200 runs. Analysis was conducted as to why the model couldn't reliably show the Alcohol Harm Paradox, and some suggestions were given for further development.

Chapter 1

Introduction

Although global alcohol consumption rates have fallen since 2010, in 2016, 43% (2.348 billion people) of the global population had consumed alcohol in the past 12 months, with the European region having the highest rate of consumption at 59.9% (449 million people) of the total population (1). Alcohol is among the leading causes of premature deaths worldwide, with alcohol being attributed to 3 million deaths per year (1). Alcohol is also the most significant risk factor for those aged 15-49 (2). This is also an age range where people are most economically active, hence the economic ramifications of alcohol-related deaths are significant.

Alcohol is well known to have both immediate and long-term adverse health effects (3). There is some conflicting literature on whether there is a safe level of alcohol consumption, with some studies showing that there is no safe level of alcohol consumption and others showing that participants with light alcohol consumption had lower risk of cancer and death than those who abstained (4; 5).

The Alcohol Harm Paradox (AHP) describes a phenomenon in which those with higher Socioeconomic Status (SES) experience less overall alcohol-associated harm even when consuming greater amounts (6). The AHP can be demonstrated both globally and locally, hence is a phenomenon which occurs across scales (1; 2; 7–9). Although the underlying causes of the AHP are still unknown, some hypotheses have been proposed as potential explanations, such as low SES individuals having less access to healthcare and heavy drinkers becoming more prone to falling down in their SES (10). The conventional approach to addressing public health issues is presupposed to changing the actions of individuals, which negates the social and economic context in which people live (11–13). These approaches have so far come up against policy resistance, suggesting an approach considering a wider societal context may be useful in understanding and solving the AHP. Fundamental Cause Theory (FCT) is a social theory which recognises the health-related ramifications of differential access to physical and social resources and has been proposed as a potential social theory to explain the AHP (13; 14).

Agent-based Modelling (ABM) is a technique used successfully in public health modelling, most notably during the COVID-19 pandemic (15). Some ABMs have been used in the context of alcohol in public health (16). This dissertation will look to build an ABM using FCT and will seek to critically analyse whether AHP can be explained through FCT.

1.1 Alcohol Harm Paradox

The AHP has emerged from a collection of literature, where the association of alcohol harms with SES was first formalised by Smith et al. in 2014 (17). Health inequalities have been apparent in almost all civilisations to date and have even been shown to emerge as a result of being low status in the animal kingdom (18; 19). The question becomes why, in a modern society with relatively widespread access to healthcare and health information, is SES still a key determining factor in someone's health? And what are the other risk factors that determine the difference in health outcomes in the context of alcohol consumption?

The Global Status Report on Alcohol 2018 by the World Health Organisation (WHO) consistently attributes SES as being a key determining factor for differential outcomes in alcohol harms. SES is a higher predictor of alcohol harm across an individual's lifetime than alcohol consumption (1). On a global level, with Europe and North America having the highest alcohol consumption rates, low and lower-middle-income countries experience the highest levels of harm related to alcohol (1). Even within the bounds of the global regions declared by the WHO, the AHP was still apparent (1). It is, therefore, appropriate to state that the AHP is a phenomenon that occurs across scales. The AHP contains many characteristics of a 'wicked problem', including the fact that policies enacted may only have better or worse outcomes and the fact the AHP can be viewed as a symptom of another problem (10; 20).

Boyd et al. conducted a systematic literature review which looked for explanations of the AHP (21). The fact this was conducted in 2022 implies that the AHP is still not well understood. Boyd notes a significant gap in the literature surrounding the AHP is the lack of research surrounding social and economic causes. A number of reports from the UK demonstrate and investigate the AHP (8–10; 22). Literature commonly divides a population by Index of Multiple Deprivation (IMD) quintiles, or Deprivation Quintile (DQ)s (8). Quintiles split the population into groups of 20%. IMD as an index, groups measures of deprivation and is a commonly used measure to quantify the deprivation of an area (23; 24).

Research by Bellis et al. acknowledges the AHP and reasons that individual-based actions such as smoking and diet choices alongside less access to healthcare resources may lead to the AHP (9). Research performed by Baldwin et al. suggested five hypotheses for the existence of the AHP (10). Baldwin then tested three of these hypotheses, including under-reporting of alcohol use, drinking patterns between SES groups and compounding effects of other health determinant factors such as diet and physical activity. Baldwin did not test differences in access to health services between different SES groups or whether the AHP could be explained by individuals with problematic alcohol habits moving into lower SES positions as a result of their consumption. Baldwin finally points towards a lack of published evidence that looks at the underlying mechanisms behind the AHP.

A theme in the literature is the lack of access to healthcare in low SES communities as a leading contributing factor behind the AHP. This can be linked to the inverse care law - first noted in 1971 and a well-known problem in public health (25). The literature to date attributes the AHP to a variety of causal mechanisms alongside differential access to healthcare. It is clear that there are multiple factors contributing to the AHP. Although multiple individual behaviours are thought to contribute towards the AHP, the literature generally misses the broader social and economic context. This is notably striking, considering

the AHP is inherently a social problem.

Figure 1.1 shows alcohol-specific deaths across Scotland between the years 2001 to 2021 (26). Figure 1.1 clearly demonstrates differential harm outcomes between those low in SES (DQ 1) when compared to those high in SES (DQ 5). As previously mentioned, differential harm outcomes are found on multiple scales, and this same pattern is observed in a global context as much as it is in a local context (1).

There is literature proposing complex systems approaches could be used to investigate health disparities, which would address the lack of social and economic context in the AHP literature (11; 14). Particularly, Boyd et al. suggest four health inequality theories could be used to investigate the AHP, one of which is FCT, which is what this dissertation will focus on modelling (14).

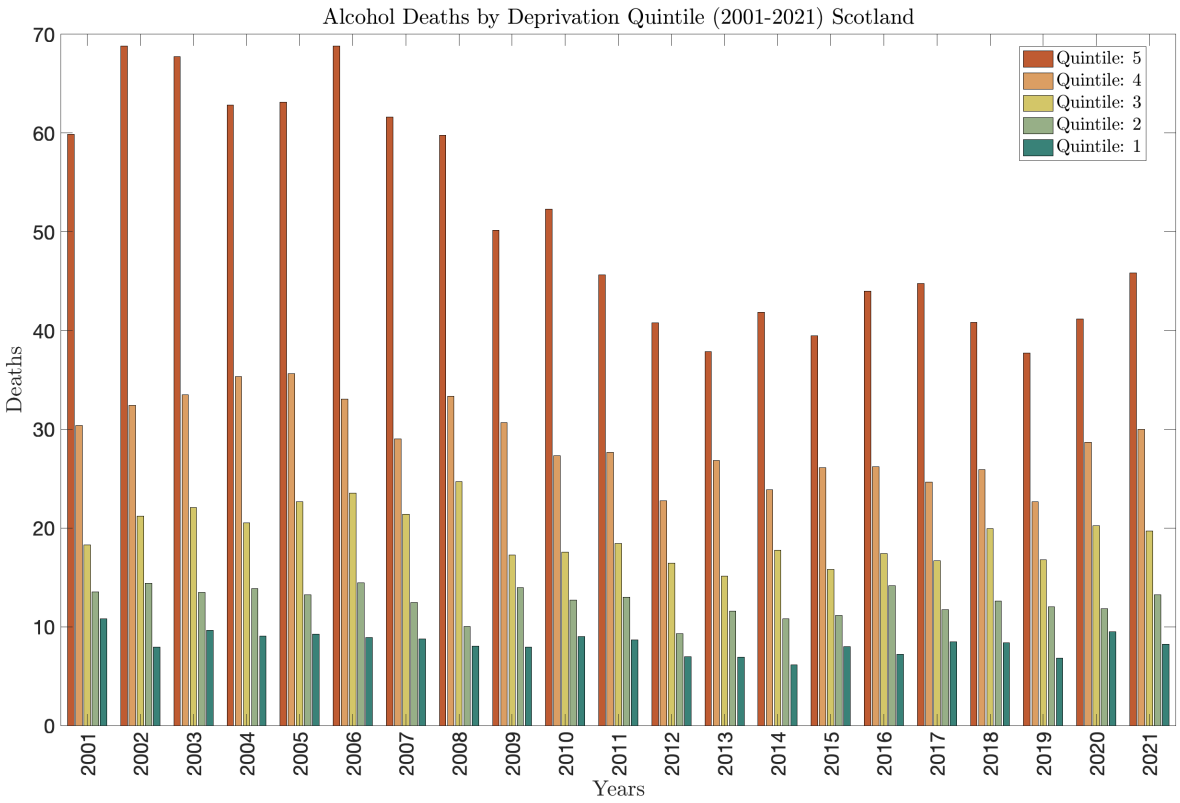


Figure 1.1: *Alcohol Specific Deaths in Scotland, divided by DQ* [1]

1.2 Fundamental Cause Theory

First noted by Link et al. FCT proposes that socioeconomic conditions are the underlying causes of the disease since low SES increases the exposure to the risk of disease (13). The precipice of FCT is that that negative health outcomes cannot be resolved solely by addressing the direct mechanisms associated with the disease itself (13). In the context of alcohol, FCT

[1] Data taken from National Records of Scotland (26)

recognises that, for example, a person with an Alcohol Use Disorder (AUD) will struggle to resolve the issue through individual action if the social and economic resources available to them are low. By contrast, someone with access to healthcare, money, knowledge, a supportive community and holds a position of high prestige in society is more likely to overcome adverse effects of AUD.

There are two ways to test whether FCT can successfully explain a phenomenon (27). First is the "preventability shifts approach", which looks at exposure to different diseases over time as well as the capacity to cure said diseases. When the capacity to cure a disease increases, if low SES individuals are seen to have a higher mortality rate, then FCT can be used to explain the phenomenon. The second is the "Manipulated preventability approach", where given a sample of a population when interventions are performed on a random set of individuals but require resources to benefit from the interventions if differential health outcomes are observed based on access to resources, then FCT can be used to explain the phenomenon.

FCT has seen success in explaining neighbourhood disadvantage and differential outcomes from both mental and physical health issues (28–30). For hard-to-cure diseases, FCT is less effective in predicting health outcomes (30). This suggests that FCT cannot explain all health outcomes in society but can be used to attribute differential health outcomes for individuals with different SES for curable diseases. Common diseases of the liver are largely curable or preventable, and liver damage is often linked to alcohol consumption. Common advice given to either prevent or aid the recovery of liver diseases is to abstain from or reduce alcohol consumption (31).

Boyd et al. (14) note that FCT is not apparent in AHP literature and argues that individual-based methods for understanding the AHP are limited as they fail to account for the socio-economic context in which people live. This was the original motivation behind the research done by Link (13). FCT seems to be useful social theory to help unravel the AHP due to the strong link between AHP and SES.

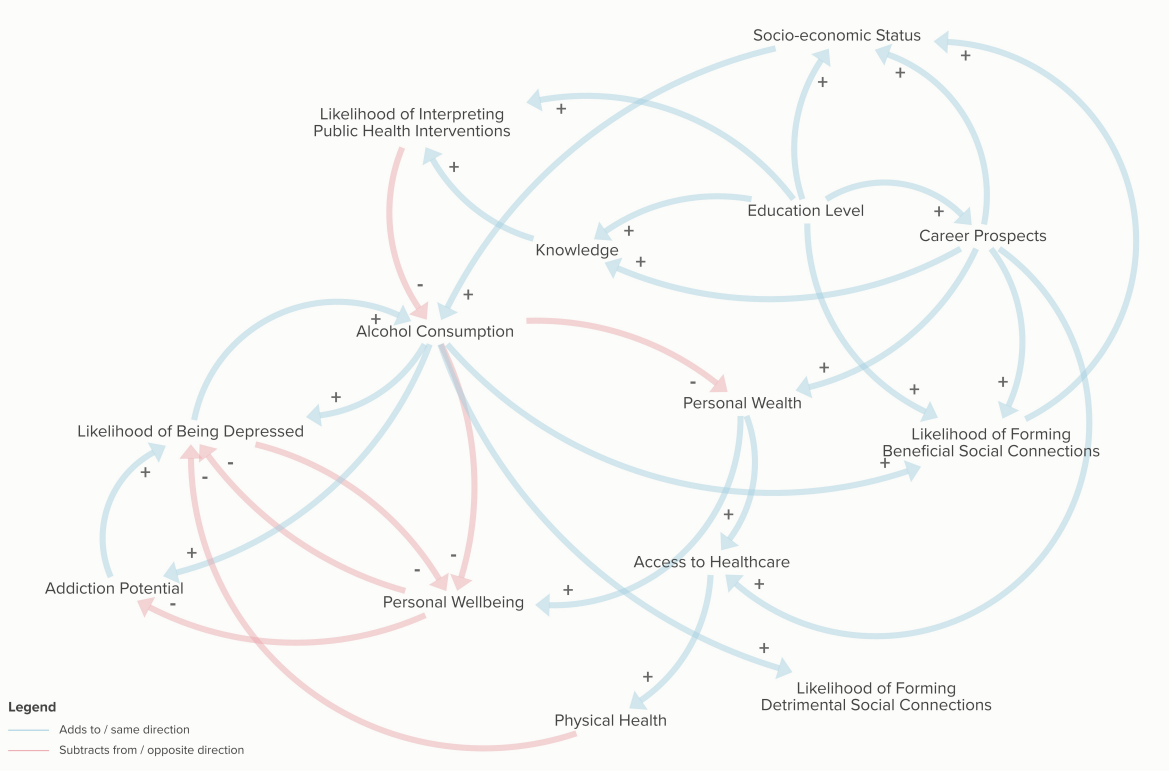


Figure 1.2: *A Causal Loop Diagram used to Visualise the AHP through FCT*

Causal loop diagrams are commonly used to form qualitative systems maps regarding the system of concern (32). A causal loop diagram, shown in Figure 1.2, was developed to place FCT in the context of alcohol consumption on an individual level. The intention was to develop an understanding of how alcohol harms developed when framing the AHP in a FCT context. Figure 1.2 shows that there are multiple reinforcing loops that link alcohol with effects that would be detrimental to the health of an individual. For example, alcohol consumption subtracts from personal well-being, personal well-being subtracts from addiction potential, addiction potential contributes to the likelihood of being depressed, and the likelihood of being depressed contributes to alcohol consumption. However, there is a reinforcing loop that contributes to what could be considered beneficial to the individual, but it relies on a high education level. Education level adds to SES, SES adds to alcohol consumption, alcohol consumption adds to the likelihood of forming beneficial social connections, and beneficial social connections add to SES. Nonetheless, there are more reinforcing loops that are detrimental to an individual's health, showing that if an individual's SES-related parameters are high, they have a higher capacity to deal with the detrimental impacts brought on by alcohol consumption.

1.3 Aims and Objectives

1.3.1 Aims

The AHP is a phenomenon in which those with higher SES experience less overall harm from alcohol consumption over their lifetime than those with lower SES, despite consuming greater quantities of alcohol. ABM has been used to try to understand the paradox within a framework informed by a social theory known as FCT. This project aims to replicate an existing ABM and critically explore the extent to which it can reproduce the AHP.

1.3.2 Objectives

Core Objectives

1. Understand the AHP, FCT and the wider context of ABMs within public health research.
2. Develop Unified Modelling Language (UML) design documents to represent the existing FCT-based ABM.
3. Use the UML design to develop a replica ABM in Repast for Python (R4P).
4. Develop a range of experiments to explore the ABM and its dynamic behaviour.
5. Implement experiments and analyse the ability of the ABM to reproduce the AHP.

Stretch Objectives

6. Critically appraise the ABM as an adequate representation of FCT within the context of AHP.
7. Propose modifications or enhancements to the ABM already in use to better represent FCT.

Chapter 2

Literature Review

2.1 Agent-based Models

ABMs are used to simulate complex systems comprising of interacting individual entities (33). ABMs are particularly of interest when the modeller seeks to understand macro-level behaviour that emerges as a result of many of these small-scale, local interactions (33). ABMs are particularly interesting in the context of social science since the individual entities are used to represent people (33). The mechanistic interaction between people then generates social phenomena as the emergent property of the model (33). Human society can be considered to be a complex adaptive system, making ABMs an appropriate modelling tool for viewing the AHP through FCT (33; 34).

Agents are generally characterised by four attributes. Agents are able to *perceive* the environment in which they reside (33). Agents are able to *perform* actions, including movement and communication with other agents (33). Agents have *memory*, whereby their previous states are recorded and impact their future states (33). Finally, individual agents each have a *policy*, a set of rules which they turn to, helping determine what next action they should take (33). The policy in this context would be FCT.

2.1.1 Models Used in Social Systems Modelling

There are numerous Complex System (CS) modelling approaches available when modelling social systems, and there exist a number of guidelines ensuring the correct modelling approach is used. The Brennan Taxonomy adapted from Breeze et al. and presented in Table 2.1, is an aggregation of 15 different CS modelling approaches and outlines the contexts in which each approach is most appropriate (35–37). Different modelling techniques capture different aspects of the system being modelled. For example, dynamic system models are best used for capturing macro-level behaviour (35). ABMs are more appropriate for understanding emergent properties that arise from many small-scale, local interactions (35).

Mechanism-based Models Mechanism-based models are rooted in Critical Realism, a philosophy in the social sciences which proposes that the objective of social science should be to understand the fundamental observed reality (38). Critical realism and emergence are closely linked. Miller asserts that 'to understand emergence, we need to identify the nature

of the disconnection between constituent parts and the whole' (38). The emergent property in the context of this dissertation is AHP, and the constituent parts are people represented as agents. Using social mechanisms to help understand macro-level social outcomes is a relatively recent transition in the social sciences (39). Mayntz and Miller both argue that searching for social 'laws' through correlation merely reveals a relationship as opposed to a causal link (38; 39). Human societies don't follow strict, time-invariant laws as with the physical systems, but mechanisms can be considered 'perceived regularities', less generally applied as with laws but still relevant when searching for emergent phenomena (39). The most famous mechanism-based model is the Coleman Boat (40), shown in Figure 2.2. This is the most relevant mechanism-based modelling approach to this dissertation since it underlies the Mechanism-Based Social Systems Modelling (MBSSM) software architecture (41). Nonetheless, there exist other mechanism-based models other than the Coleman Boat, for example, the Aggregation Model seen in Figure 2.1.

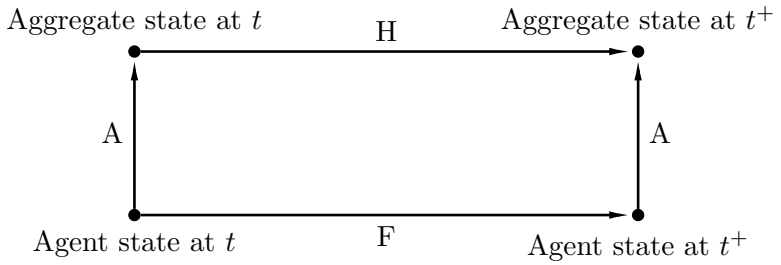


Figure 2.1: *A Representation of an Aggregation Diagram* [1]

There exists some critique of mechanism-based modelling, arguing that some social systems are adequately explained without the use of mechanism-based models (43). Nonetheless, the uptake of mechanism-based models in the social sciences has facilitated a more concrete understanding of the underlying causal mechanisms that lead to emergent social phenomena (43). This makes a mechanism-based approach appropriate for helping unravel the AHP.

Dynamic System Models Modelling social systems as dynamic models involves taking sociological concepts, turning them into variables, and translating the interrelationships between the concepts into a dynamic systems model (44). Applying a dynamic systems approach to a social system starts with outlining causal relationships between the variables of interest and mapping these interrelationships as a causal loop diagram, similar to what is seen in Figure 1.2 (45). This technique was utilised by Salmon et al. to model drink-driving-related trauma prevention (45). In the application of dynamic system modelling, Salmon was able to effectively model the impacts of policy interventions to reduce drink-driving-related road trauma (45). Salmon identifies that the key strength of a system dynamics model is to understand the dynamics of whole system behaviour over time (45). Salmon also identifies some limitations. Since dynamic systems models rely on the opinion of subject matter experts, this presents an issue of validity as well as making the development procedure time-consuming (45).

[1] Adapted from Hédoin, 2013 (42).

			A	B	C	D
			Cohort/aggregate-level/counts		Individual-level	
			Expected value, continuous state, deterministic	Markovian, discreet state, stochastic	Markovian, discrete date	Non-markovian, discrete state
1	No interaction	Untimed	Decision tree rollback or comparative risk assessment	Simulation decision tree or comparative risk assessment	Individual sampling model. Simulated patient-level decision tree or comparative risk assessment	
2	”	Timed	Markov Model deterministic	Simulation Markov Model	Individual sampling model: Simulated patient-level Markov model	
3	Interaction between entity and environment	Discrete Time	System dynamics (finite difference equation)	Discrete Markov chain model	Discrete-time individual event history model	Discrete-time discrete event simulation
4	”	Continuous Time	Systems dynamics (ordinary differential equations)	Continuous time Markov chain model	Continuous time individual event history model	Continuous-time discrete event simulation
5	Interaction between heterogenous entities/spatial aspects important		X	X	X	Agent-based simulation

Table 2.1: *Table Based on the Brennan Taxonomy* [2]

[2]Adapted from Breeze et al., 2023. Original Adaptation by Briggs et al. (35; 46)

2.1.2 Complex Systems and Public Health

Complexity science, in general, has much to offer the healthcare sector. Measures of the complexity of bodily functions have been found to be more accurate than traditional methods, such as heart rate and breathing rate, for measuring organismic function (47). Reduction in complexity can be used as a more effective measure for assessing the impact of disease (47). On the macro level, Rutter et al. argue that many approaches to public health are based on reductionist, linear interpretations of relationships between policies and individuals (48). Where randomised controlled trials are used to inform policy, often the policies encounter policy resistance when applied to the population at large (48). ABMs give researchers and policymakers the capacity to model how *local interactions* lead to the emergence of complex, large-scale behaviour (33; 49). ABMs have had success in explaining social phenomena such as segregation (50). One of the more recent and well-known applications of agent-based modelling in public health was during the COVID-19 pandemic, where ABMs were used to evaluate and justify stay-at-home orders (15).

2.1.3 Agent-based Models & Alcohol Policy

A scoping literature review was conducted by McGill et al. which looked for literature using used complex systems approaches to explore alcohol-related harms, some of which included ABMs (16). Some ABMs focus on alcohol consumption among young and college-aged people (51–55), whilst others investigate more general populations (56–62). Certain ABMs focus on alcohol consumption during events where alcohol is consumed, such as nights out or parties, with the intention of finding ways to reduce alcohol-related violence (59–66). There are also some ABMs prioritising intervention strategies (51; 53; 55; 57; 59; 62; 63; 67) and some prioritising understanding behavioural mechanisms (51–54; 56; 58–60; 64–66). There is notable overlap between ABMs which look at behaviour and intervention strategies, as often, understanding of behaviour is needed before the appropriate strategies can be employed. The models which are based on real-world data are initialised based on data sets from the US, UK, The Netherlands and Australia, hence there is a bias towards analysing populations from more advanced economies.

A common theme in the literature is to help shape types of interventions to curb the negative effects of alcohol by attempting to outline the mechanisms related to problematic alcohol consumption. Many of these models focus on the direct harms associated with events related to alcohol consumption, such as alcohol-related violence, trips and falls and vehicle-related accidents. No ABMs mentioned in McGill's review focus on SES or the AHP. This may suggest that ABM as a technique may not be suitable for investigating the AHP, but other computational complex systems approaches, such as dynamic systems modelling, fail to capture the complexity involved where there are multiple interacting agents such as in a population (35).

2.1.4 Designing Agent-based Models

UML is an approach to software design which presents many benefits in the development of ABMs (68). Bersini argues the case for class, state, and sequence diagrams for the effective implementation of an ABM in software (68).

Class diagrams, which describe the behaviour of and relationship between entities in a model, are the most important models to design in the development of ABMs (68). Class diagrams comprise entities and relationships between those entities. Entities take the form of classes; the relationship between entities can take many forms, such as inheritance and association (68; 69). Classes represent objects in software, a data structure containing attributes (variables) and methods (functions) (69). The relationships between classes then describe how methods and functions relate between the two connected objects (69). Certain structures comprising classes and the relationship between them give rise to what are known as design patterns (68). The mediator design pattern is a design pattern used to mediate the interaction between different objects (70).

Sequence Diagrams describe how classes or objects interact with each other over time (68). Sequence diagrams frame time vertically and allow the designer to conceptualise how objects in loops communicate (68).

Finally, state diagrams are used to visualise and design a process that an object is due to go through, where the state diagram allows for the visualisation of agent states as the agent passes through the process (68).

2.1.5 Mechanism-based Social Systems Modelling

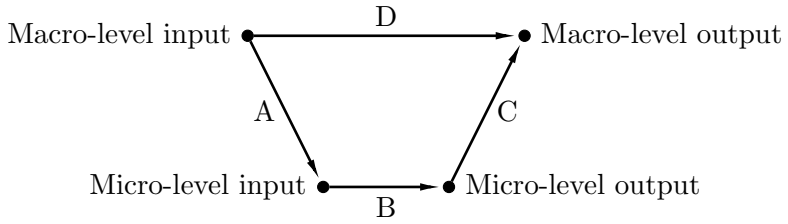


Figure 2.2: *A representation of the Coleman Boat as adapted by Hedstrom and Swedberg* [3]

MBSSM is a software architecture developed by Vu et al. which uses a mechanistic approach to modelling social systems. The framework was developed to assist the creation of ABMs intended to model the interaction between a social systems' larger-order structures with individual-level behaviours, known as macro and micro elements respectively (41). The Coleman Boat seen in Figure 2.2 is an abstraction of interactions between a social system's macro and micro elements which forms the basis of MBSSM (40). Hedstrom and Swedberg narrowed down the abstraction and described how the Coleman Boat could be applied when modelling a social system from a mechanistic perspective (71).

Hedstrom and Swedberg coined three key terms relating to the Coleman Boat and the interaction between the macro and micro levels (71). First, 'situational mechanisms' relate to the interactions between the macro-level input and the micro-level input, part A in Figure 2.2 (71). Situational mechanisms can be thought of as how an institution interacts with the people inside said institution. Second, 'action mechanisms' refer to the interaction between micro-level input and micro-level output of the Coleman Boat, part B in Figure 2.2

[3] Adapted from Hedstrom and Swedberg 1996 (71).

(71). Action mechanisms can be thought of as either individuals acting on themselves (ie. drinking alcohol) or individuals interacting with other individuals (ie. sharing information). Finally, 'transformational mechanisms' refer to the interaction between micro-level output and macro-level output, part C in Figure 2.2 (71). Transformational mechanisms can be thought of as either the simple or complex aggregation of individual actions. In other words, transformational mechanisms are changes to the macro-level of the system, brought on by the mechanistic actions of individuals. Hedstrom and Swedberg didn't propose a term for macro-macro interactions, asserting that mechanistic analyses of social systems primarily focus on the general macro-micro-macro schema (71).

Hedstrom and Swedberg propose three key principles for the mechanistic analysis of social systems: direct causality, limited scope, and methodological individualism (71). The direct causality principle states that researchers should strive to look for direct relationships between cause and effect, with Hedstrom and Swedberg arguing that mechanism-based explanations of social systems should strive for 'tight coupling' between said cause and effect (71). The second principle, limited scope, explains that sociology should not strive for universal explanations of social phenomena, hence mechanistic models should strive to explain a number of mechanisms only appropriate for the phenomenon being investigated (71). Finally, Hedstrom and Swedberg argue that association between variables in a social system do not lead to macro-level outcomes, but rather mechanistic actions of individuals towards themselves and other individuals, and the various consequences of this lead to macro-level outcomes (71). The three principles laid out by Hedstrom and Swedberg make their interpretation of the Coleman Boat a strong candidate for assisting the design of an ABM around a social theory. The Coleman Boat and the situational, action, and transformational mechanisms outlined by Hedstrom and Swedberg form the basis of MBSSM (41).

There exist a number of other mechanistic approaches for modelling social systems, Tilly breaks down a mechanistic approach into three subcategories: environmental, cognitive, and relational (72). Each of the three respective subcategories can be attributed to the Coleman Boat concept as macro-micro, micro-micro, and micro-macro, respectively (72). Similarly, Archer proposes a similar cyclic approach, focusing on the interaction between the individual and the collective (73). Finally, Sawyer proposes a cyclic, 5-level approach to social systems modelling (74). The 5-level approach used by Sawyer results in less distinction between the individual and the collective, which may be advantageous or disadvantageous depending on the context of the social system being analysed (74). Nonetheless, these approaches share a commonality in that they represent the interaction between larger-order social structures, such as governments or institutions, with granular entities, such as individuals or households (41). Despite the discrepancies between other mechanistic approaches, MBSSM is designed to be flexible enough to be altered to facilitate the appropriate mechanistic approach for the given problem. This is done by putting a core emphasis on the macro-micro, micro-micro and micro-macro aspects of MBSSM in the framework whilst still allowing the developer to implement macro-macro interactions should they deem it necessary (41).

MBSSM Topology

MBSSM takes the core principles of mechanism-based modelling and presents a software architecture with which to build an ABM through the implementation of social mechanisms (41). MBSSM uses terminology developed by Hedstrom and Swedberg to specify the mech-

anistic macro-micro-macro interactions: situational mechanisms, action mechanisms, and transformational mechanisms (41).

Situational Mechanisms Situational mechanisms outline the macro-micro relationship in the model and represent the relationship between the agents as a collective, usually through the implementation of an agent network or a shared spatial domain (41). Since situational mechanisms relate to a spatial domain, agents have the capacity to react to changes in the spatial domain (41).

A real-world example of changes to individuals based on collective behaviour might be the introduction of a viral trend - people see the trend on social media and alter their own behaviour in that they replicate it. Similarly, a change in the transport infrastructure of a city may impact whether or not a person decides to drive or take public transport, in essence altering their individual behaviour. In essence, situational mechanisms describe how the wider society of agents and the location in which that society finds itself interact with individuals.

Action Mechanisms Action mechanisms outline the micro-micro relationship in the model. In the context of MBSSM, action mechanisms represent changes to an individual's decisions, whether conscious or unconscious (41). In a real-world context, an action mechanism is an individual relating information passed to them through social media to their own life (i.e. thinking about information). Similarly, an action could be considered an individual communicating this same information to another individual.

Transformational Mechanisms Transformational Mechanisms represent the conglomeration of actions of individuals that lead to changes to the collective population or their spatial domain (41). In a real-world context, transformational mechanisms can be thought of as the outcome of an election. The emergence of a social system can be thought of as the outcome of many iterations of transformation mechanisms.

2.2 Summary

A mechanistic approach to modelling social systems allows social scientists to explore how emergent behaviour arises from small-scale, mechanistic interactions between autonomous individuals. Complex systems modelling approaches are becoming increasingly relevant in both fields since the reductionist, linear attribution of causal relationships doesn't adequately explain how interactions between human beings lead to large-scale societal outcomes. ABM is a technique enabling the simulation of social systems, allowing social scientists and public health experts to explore how emergent properties of the modelled system change with time.

Boyd et al. state that with the advent of MBSSM twinned with the interest of policymakers to use ABM-based tools, it is an appropriate time to investigate the AHP through ABMs (21; 75) Summarising, there is a clear gap in the AHP literature, which has failed to find the underlying causes of the AHP due to the reliance on individual-based interventions. FCT presents a new theory to investigate the AHP, proposing that SES and the difficulties and lack of resources that accompany being low SES lead to the differential outcomes in alcohol harms described by the AHP. ABMs have seen success in public health modelling and in finding intervention strategies for dealing with alcohol-related harms, and ABM seems like an appropriate technique to investigate whether AHP can be explained through FCT, especially since the development of the MBSSM framework.

Chapter 3

Model Development

3.1 Conceptual Modelling

Experiment replication in the physical sciences is a well-established practice. Wilensky and Rand argue that the replication of computer models is more important in the context of computational modelling as not only does it facilitate the replication of the original phenomenon, but it contributes to the validation and verification of the original model, as well as contributing toward a common method of simulating the original phenomenon, useful for other modellers (76). Wilensky and Rand emphasise the importance of attempting model replication before accessing original source code (76). Section 3.1 formalises important concepts from the ODD laid out by Boyd, which are then used to inspire software design, laid out in Section 3.2 (77).

The basis of the MBSSM framework is the Coleman Boat, outlined in Figure 2.2 (41). Therefore, the Coleman Boat is re-framed in the context of FCT, outlining situational mechanisms, action mechanisms, and transformational mechanisms.

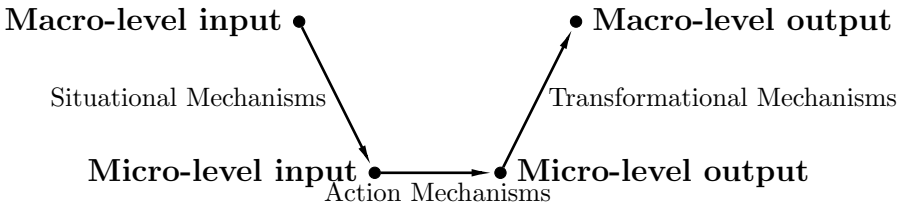


Figure 3.1: *A representation of the Coleman Boat in the context of FCT* [1]

3.1.1 Overview

FCT was placed into a mechanistic context by Boyd, who identified two macro-level entities: an environment and a communicator (77). The communicator entity represents an abstraction of "the source of innovation regarding risk, treatment, and prevention" of disease, hereafter referred to as 'the communicator', 'communicator entity', or 'communicator' (77). The environment entity represents the spatial domain in which agents reside, hereafter

[1] Adapted from Vu et al. (41)

referred to as 'the environment', 'environment entity', or 'environment' (77). Finally, Boyd identified that micro-level entities should be considered as individual people (77), hereafter referred to as 'micro-agent(s)' or 'agent(s)'. Finally, Boyd determined that an appropriate time step in the context of the simulation would be one week, meaning $k = 1$ week.

3.1.2 Agents

Agents are defined following the format outlined by Purshouse in describing a cellular automaton (78).

1. State vector $\vec{x}_n$
2. Starting state $\vec{x}_0$
3. Input vector $\vec{u}_m$
4. State transition function $\delta : \vec{x} \times \vec{u} \rightarrow \vec{x}^{\,+}$

Let the set of all agents be $\mathcal{A}$, for each individual micro-agent $a_i \in \mathcal{A} \; \forall \; i \in [1, N] \in \mathbb{Z}$, where N is the number of agents. Each individual agent has an associated state vector $\vec{x}_n$, for $n = 10$ states. The description of each state can be seen in Table 3.1. Each agent has a unique starting state vector $\vec{x}_0 = \vec{x}(k = 0)$. Similarly, each agent has a vector of inputs $\vec{u}_m(k)$ of size $m = 1$ corresponding to the number of inputs (78). Each agent possesses a state transition function δ, used to map inputs from $\vec{u}$ to the next state $\vec{x}^{\,+}$.

State Variables

The state variables in the simulation correspond to agent attributes that change with time. Some state variables rely on parameterised variables, found in Table 4.1 of the report.

State Variable	Variable Name	Description
x_1	Age	$x_1 \in [\phi_1, \phi_2] \in \mathbb{Z}$
x_2	DQ	$x_2 \in [1, 5] \in \mathbb{Z}$
x_3	Mean Weekly Units	$x_3 \in [0, 130] \in \mathbb{R}$
x_4	Drinking Status	$x_4 \in [0, 5] \in \mathbb{Z}$
x_5	Desired Connections	$x_5 = 3$
x_6	Power	$x_6 \in [0, 1] \in \mathbb{R}$
x_7	Prestige	$x_7 \in [0, 1] \in \mathbb{R}$
x_8	Personal Wealth	$x_8 \in [0, 1] \in \mathbb{R}$
x_9	Knowledge	$x_9 \in [0, 1] \in \mathbb{R}$
x_{10}	Communicated Information	$x_{10} = \mathbb{C} \in \mathbb{Z}$

Table 3.1: *State Variables $\vec{x}$ Based on Dynamic Agent Attributes* [2]

The state vector explicitly describes agent states that change with time k, so constant agent attributes have been omitted in this context, and are defined as part of state transition functions.

[2] Adapted from Boyd (77)

Agent Inputs

An agent a_i can only have one possible input, the communication of information I from either the communicator: $I(\mathcal{C})$ or another agent $I(a_j)$.

Network

Agents are networked and can interact. The graph $G(\mathbb{V}, \mathbb{E})$ represents the agent network, the set of all vertices $\mathbb{V}(G)$ and the set of all edges $\mathbb{E}(G)$. Both the communicator and agents are vertices in the graph, that is $\mathcal{A} \wedge \mathcal{C} \in \mathbb{V}(G)$. Connections between vertices are weighted, with edges defined as $\mathbb{E} \subset [V]^2 \times \mathbb{R}$ giving $\mathbb{E}$ as a set of ordered triples. Let $\omega : \mathbb{E} \to [0, 1] \in \mathbb{R}$, giving each edge defined as $e(a_i, a_j, \omega)$. The communicator entity $\mathcal{C}$ is a unique vertex in the context of the simulation since its only role is to generate and broadcast information, denoted $I(\mathcal{C})$, to other agents. Hence, the communicator can be considered to have an out connection to all micro-agents $\exists\ e(\mathcal{C}, a_i, 1)\ \forall\ i \in [1, N]$, where $e \in \mathbb{E}(G)$. The number of vertices and edges in the graph remains constant through each simulation, that is $|G|(k) \wedge ||G||(k) = |G|(k^+) \wedge ||G||(k^+)$.

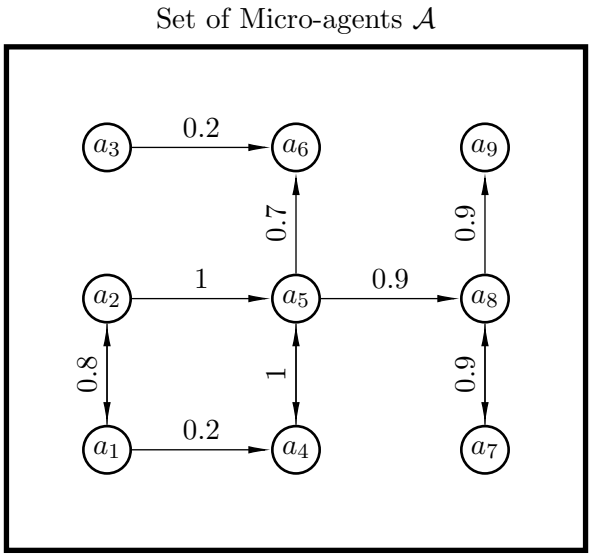


Figure 3.2: *An Example Subset of Networked Micro-agents $\mathcal{A}$*

Other than being connected to the communicator, agents are connected based on the similarities between their personal and FCT-based characteristics, such as deprivation quintile, age, sex, and drinking status (77). Formally, these connections are written $\mathbb{E}(G) = e(a_i)$. Each edge weight is determined by a discrete uniform distribution: $\omega(a_i, a_j) = U_{(0,1;0.1)}$, formed at the start of the simulation. Figure 3.2 gives a simplified representation of the agent network.

3.1.3 Situational Mechanisms

The situational mechanisms in the simulation represent interactions between the state (macro-level input) and the agents (micro-level input). In the context of alcohol policy, this relates

to intervention strategies generally through the spreading of information. Situational mechanisms in the model can be considered healthcare campaigns and other information-focused intervention strategies; these are the most prevalent types of strategies employed by central and local governments (14; 79).

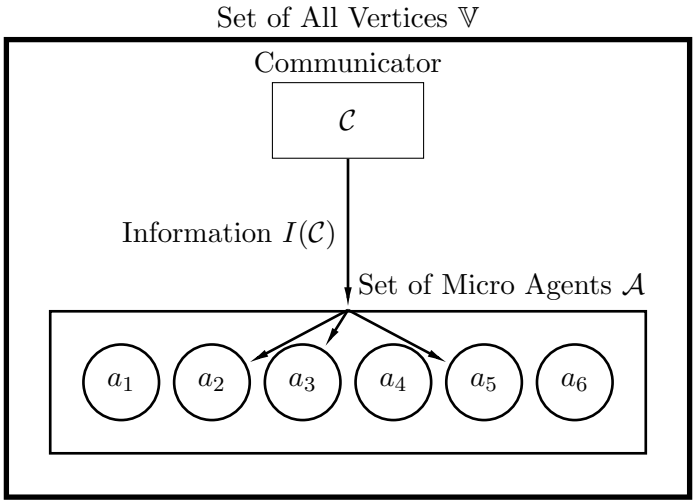


Figure 3.3: *An Abstracted Representation of Interaction Between the Communicator Agent and the Set of Micro Agents*

Risk-related information $I(\mathcal{C}(k))$ is broadcast to a random subset of agents $\mathcal{A}_\mathcal{C}$ from the communicator $\mathcal{C}$ at each time step k, forming the link between macro-level inputs and micro-level inputs (77). Using Figure 3.3 as an example, the subset of micro-agents receiving information would be $\mathcal{A}_\mathcal{C} = [a_2, a_3, a_5]$. $\mathcal{A}_\mathcal{C}$ is randomly generated at each time step k hence $\mathcal{A}_\mathcal{C}(k) \neq \mathcal{A}_\mathcal{C}(k^+)$. Nonetheless, the number of agents, that is, the size of the set of agents receiving communication, remains the same through each simulation: $|\mathcal{A}_\mathcal{C}|(k) = |\mathcal{A}_\mathcal{C}|(k^+) \; \forall \; k$. The change is in *which* agents receive the communicated information. In the case of Figure 3.3, $\mathcal{A}_\mathcal{C} = [a_2, a_3, a_5]$. Figure 3.3 shows the interaction of the communicator with micro agents, where the communicator broadcasts information that the micro agents then decode as part of the action mechanisms (77).

3.1.4 Action Mechanisms

Action mechanisms are characterised by changes in the agent states as well as communication between agents (77). Since FCT purposes that differential access to resources such as knowledge, personal wealth, power, community, and healthcare results in differential outcomes for curable diseases, action mechanisms must relate to these aspects of FCT for each individual agent (13; 77).

Solving and Communicating Information

Agents store communicated information $I(\mathcal{C})$ in the set of communicated information $\mathbb{C} = x_{10}$. Each piece of information communicated is in the form of a random, fixed-length binary string $I(\mathcal{C}) \equiv \log_2(2^4)$, for a fixed binary number of length 4. Information and 'event'

are used synonymously in the report. The set of communicated information contains two subsets of solved ($\mathbb{S}$) and unsolved ($\mathbb{U}$) information, hence $\mathbb{S} \wedge \mathbb{U} \subset \mathbb{C}$. Every tick, agents attempt to adapt to one of the communicated pieces of information, moving that information from $\mathbb{U}$ to $\mathbb{S}$ if successful. The overall goal for each agent is to adapt to every piece of information communicated to them. Hence, the target state for every agent in the simulation is $\mathbb{S}(k_f) = \mathbb{C};\ \mathbb{U}(k_f) = \emptyset$.

Agents attempt to adapt to communicated information $I(\mathcal{C})$ by 'employing' the FCT-related resources they have available (77). Resources are employed by taking the sum of all FCT-related resources(77). If the sum of all FCT-related resources is greater than the *decimal value* of the information communicated, agents are said to have adapted to the information communicated to them, outlined in Equation 3.1 (77). Since it costs FCT-related resources to adapt to information, agents 'pay' for both successful and unsuccessful adaptations ϕ_4 and ϕ_5 respectively (77). If agents successfully adapt to information, they receive a knowledge benefit ϕ_6 (77).

$$I(\mathcal{C}) \leq \sum_{i=6}^{9} x_i(k) \implies \text{Agent Adaptation} \tag{3.1}$$

Agents have the capacity to communicate information that they have solved to another agent in their personal network. If agents are communicated information from another agent, that is, a_i receives information $I(a_j)$from a_j, a uniform distribution $U_{(0,1)}$ determines whether the communication is successful and becomes a member of the set of solved events $I(a_j) \in \mathbb{S}$. If the communication is unsuccessful, then the information becomes a member of the set of unsolved events, $I(a_j) \in \mathbb{U}$. The threshold requirement for the communication to be successful is a parameterised value ϕ_{12}, seen in Table 4.1.

$$\begin{cases} I(a_j) \in \mathbb{S} & \text{for } U_{0,1} \leq \phi_{13} \\ I(a_j) \in \mathbb{U} & \text{otherwise} \end{cases} \tag{3.2}$$

The intention behind the approach in Equation 3.2 is to reflect how some communicated information between two individuals is well understood by the recipient, whilst other times it is not. This approach differs from the inter-agent communication approach used by Boyd, where if information were communicated, then the receiving agent (in this case a_i) would have their FCT parameters scaled when attempting to adapt to the communicated information(77). Agent adaptation to information results in changes to agent FCT-based parameters. Upon a successful adaptation, where $I(\cdot) \in \mathbb{S}$, individual agents receive a knowledge benefit, determined by a parameterised value ϕ_6, seen in Table 4.1. Any time an agent attempts to adapt, they must employ or use up FCT-related resources. As such, whether for a successful or unsuccessful adaptation, agents must 'pay' an adaptation cost, where the parameters ϕ_4 and ϕ_5 represent successful and unsuccessful adaptation costs, respectively. As with other parameterised values, these are found in Table 4.1.

3.1.5 Transformational Mechanisms

FCT considers an individual's deprivation level as one of the core reasons for differential access to healthcare (13). A common practice for identifying the deprivation of a given area is by using IMD quintiles, henceforth referred to as Deprivation Quintile(s) (DQ) (24). IMD

quintiles are represented by locations in the environment in which the agents exist. Transformational mechanisms in the simulation are characterised by changes to agent locations, where agents have the chance to move up or down IMD quintiles on a yearly basis (77).

Deprivation Swap

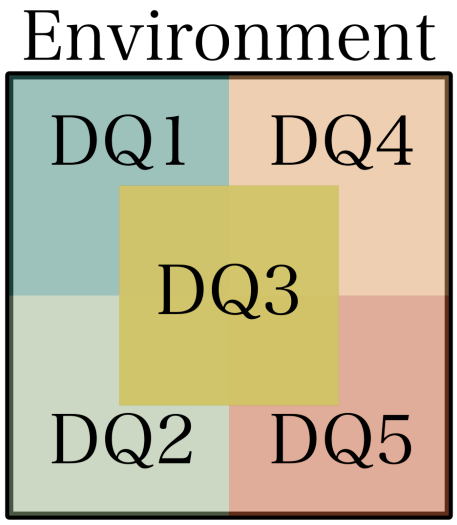


Figure 3.4: *Representation of Deprivation Quintiles in the Simulation*

Figure 3.4 shows the environment in which agents reside. The matrix of all possible locations in the environment is given as $B \in \mathbb{Z}^{50\times 50}$. The environment can be thought of as a 50×50 chess board, where a single agent can only be in one location on the board at a time. Agents are restricted to reside only in locations that correspond to their DQ, x_2. Agents attempt to swap on a yearly basis; every 52 ticks $k \equiv \mod(52)$. Agents swap deprivation quintiles according to the deprivation swap model employed by Boyd and developed by Giorgi et al. (80). The method outlined by Giorgi et al. describes how agents below age 30 and in less deprived DQs are said to be more likely to move to a higher DQ, corresponding to moving down the social hierarchy (80). Similarly, agents above age 30 and in a high DQ are said to be more likely to move to a lower DQ, corresponding to moving up the social hierarchy (80). Georgi et al. comment that this is just an example of a deprivation swap model and other deprivation swap models could be used (80). As such, whether an agent is able to swap up or down DQs is determined by Equation 3.3.

$$\text{Swap Type} = \begin{cases} \text{Down} & \text{if } x_1 < 30 \\ \text{Up} & \text{if } x_1 \geq 30 \end{cases} \tag{3.3}$$

Table 3.2 refers to the different probabilities an agent will swap up or down DQs depending on their age, x_1. When the agent's age and DQ are determined, their swap probability is put against a random uniform distribution $U_{(0,1)}$ to determine whether or not that agent will

swap. To give a clearer explanation of the swapping process, consider two agents, a_i and a_j. Let the age and DQ of a_i be $x_{i,1} = 25;\ x_i,2 = 1$. Similarly, let the age and DQ be $x_{j,1} = 40;\ x_{j,2} = 5$. The example assumes that both a_i and a_j have met the requirements to swap, that is $U_{(0,1)} \leq P_s(a_i) \wedge P_s(a_j)$. Let the set $\mathcal{S}_u$ and $\mathcal{S}_d$ be the swap up and swap down sets, respectively. Hence, agent a_i becomes a member of $\mathcal{S}_d$ and a_j becomes a member of $\mathcal{S}_u$, that is $a_i \in \mathcal{S}_d;\ a_j \in \mathcal{S}_u$. Agents form pairs to swap between DQs, denoted $\mathcal{P}(\mathcal{S}_u, \mathcal{S}_d)$. To maintain an equal number of agents in each quintile, there is a hard requirement such that $|\mathcal{S}_u| = |\mathcal{S}_d|$, and more strictly, that the number of agents exiting a DQ must equal the same amount of agents entering that DQ, that is $\mathcal{P}(\mathcal{S}_u, \mathcal{S}_d) = \text{constant}\ \forall \mathcal{S}_u, \mathcal{S}_d$. In the case of the agents a_i and a_j, $|\mathcal{S}_u| = |\mathcal{S}_d| \iff 1 = 1$ and $\mathcal{P}(\mathcal{S}_u, \mathcal{S}_d) = (a_j, a_i)$. The agents are then assigned each others' deprivation quintile values, hence $x_{i,2} \coloneqq x_{j,2};\ x_{j,2} \coloneqq x_{i,2}$. Once the swap is completed, the swapping sets become emptied, such that the same agents are not swapped during the next swapping event $\mathcal{S}_u, \mathcal{S}_d \coloneqq \emptyset$. Agents subsequently move to a random area that corresponds to their deprivation quintile on the board B.

Swap Type	DQ1	DQ2	DQ3	DQ4	DQ5
Down	$P_s(a) = 0.3$	$P_s(a) = 0.3$	$P_s(a) = 0.4$	$P_s(a) = 0$	$P_s(a) = 0$
Up	$P_s(a) = 0$	$P_s(a) = 0$	$P_s(a) = 0$	$P_s(a) = 0.4$	$P_s(a) = 0.6$

Table 3.2: *Deprivation Swap Table*[3]

Alcohol Consumption and Risk

The consumption of alcohol, both in the short and long term, carries an aspect of risk (4). In the model, the risk associated with alcohol consumption is calculated on a yearly basis $k \equiv \text{mod}\ (52)$ (77). The absolute risk of an agent, as a function of mean weekly consumption c, is outlined in Equation 3.4 (77).

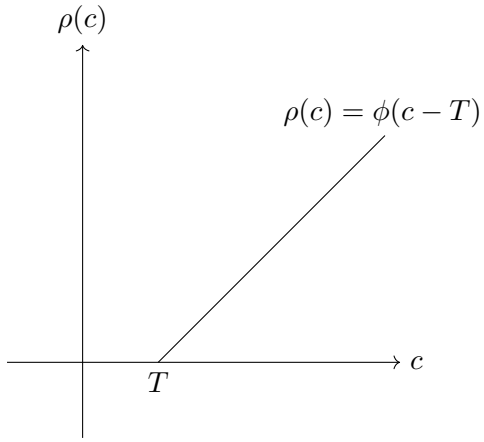


Figure 3.5: *Absolute Risk ρ as a Function of Alcohol Consumption c.* [4]

[3] Adapted from Georgi et al. (80)

Figure 3.5 demonstrates the absolute risk function used to relate alcohol consumption and agent risk in the model, and is explained through equation 3.4.

$$\rho(c) = 100 \cdot \phi \cdot (c - 0.5)$$

where

$$\phi = \begin{cases} \phi_8, & \text{for males} \\ \phi_9, & \text{for females} \end{cases} \tag{3.4}$$

Equation 3.5 refers to the modified risk function, which takes into account agent age x_1 as well as consumption c, total communicated information $\mathbb{C}$, total unsolved information $\mathbb{U}$, and age in the context of a logistic function. The modified risk function is used to calculate the risk of death. The variable ϕ_{10} is a β risk modifier and is included in the original model by Boyd (77). The original values used by Boyd are $\phi_8 = 0.00389$ and $\phi_9 = 0.00101$ for male and female agents, respectively (77).

$$\bar{\rho}(c, x_1) = \rho(c) \cdot (1 - \frac{|\mathbb{U}|}{|\mathbb{C}|} - 0.5)^{\phi_{10}} \cdot \frac{1}{1 + e^{-\frac{x_1 - 45}{9.5}}} \tag{3.5}$$

Death in the context of the simulation corresponds to assigning agent age to the minimum age of the simulation, a parameterised value $x_1 \coloneqq \phi_1$.

3.1.6 State Transformation Functions

The state transformation function

State Transition Function δ	Variable Name	Description
$x_1(k^+) = x_1(k) + 1$ if $k \bmod 52 = 0$	Age	$x_1 \in [\phi_1, \phi_2] \in \mathbb{Z}$
$x_2(k^+) = swap(a_i, a_j)$ See Section 3.1.5	DQ	$x_2 \in [1, 5] \in \mathbb{Z}$
$x_3(k^+) = x_3(k)$	Mean Weekly Units	$x_3 \in [0, 130] \in \mathbb{R}$
$x_4(k^+) = x_4(k)$	Drinking Status	$x_4 \in [0, 5] \in \mathbb{Z}$
$x_5(k^+) = x_5(k)$	Desired Connections	$x_5 = 3$
$x_6(k^+) = x_6(k) - \frac{x_6 \cdot \phi_{4,5}}{4}$	Power	$x_6 \in [0, 1] \in \mathbb{R}$
$x_7(k^+) = x_7(k) - \frac{x_7 \cdot \phi_{4,5}}{4}$	Prestige	$x_7 \in [0, 1] \in \mathbb{R}$
$x_8(k^+) = x_8(k) - \frac{x_8 \cdot \phi_{4,5}}{4}$	Personal Wealth	$x_8 \in [0, 1] \in \mathbb{R}$
$x_9(k^+) = x_9(k) - \frac{x_9 \cdot \phi_{4,5}}{4} + \phi_6$	Knowledge	$x_9 \in [0, 1] \in \mathbb{R}$
$I(\cdot(k)) \in x_{10}(k^+) = \mathbb{C}(k^+)$	Communicated Information	$x_{10} = \mathbb{C} \in \mathbb{Z}$

Table 3.3: *State Transition Functions Based on Dynamic Agent Attributes*

[4]Adapted from Boyd, 2023 (77).

3.2 Software Design and Implementation

3.2.1 Full UML Class Diagram

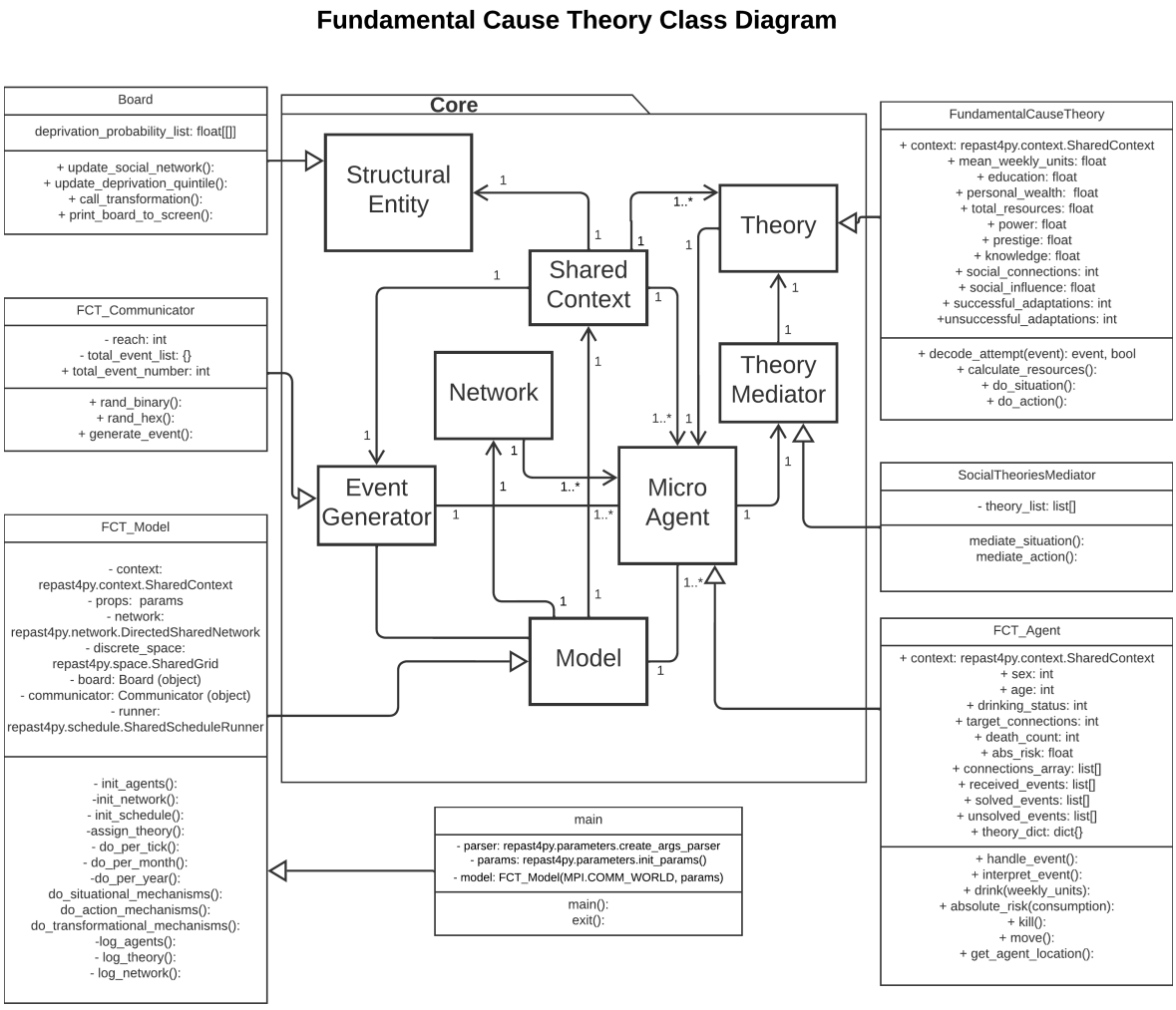


Figure 3.6: *Full UML Class Diagram*

The overall model architecture takes the design principles laid out by Vu et al. and implements MBSSM into an FCT context, building on the design laid out by Boyd (41; 77). The original MBSSM architecture can be found in Appendix B, Figure B.1. There are a number of similarities between Figure 3.6 and the architecture presented by Vu, namely the mediator design pattern utilised between the `MicroAgent`, `Theory`, and `TheoryMediator`; and the link between the `Model` and `MicroAgent` abstract classes. Similarly, the usage of abstract classes and the inheritance relationship between the abstract class and the instantiated classes remains largely the same.

Nonetheless, there are some key differences between the model architecture presented in Figure 3.6 and the architecture used by Vu. The introduction of an `EventGenerator` abstract

class and the `FCT_Communicator` are used to implement the situational mechanisms outlined in Section 3.1.3. The removal of the `StructuralMediator` was taken in tandem with the decision to implement the `FCT_Communicator` and `EventGenerator` as separate structural entities, despite them both being macro-level entities. In the context of the Coleman Boat, the `StructuralEntity` in Figure 3.6 manages the transformational mechanisms, whilst the `EventGenerator` manages the situational mechanisms. As such, there was no need for mediation or regulation between the `FCT_Communicator` and the `Structural Entity`. Finally, to manage the interactions between agents, a `Network` object was implemented.

Software Implementation

The model was implemented using R4P, a Python adaptation of the REPAST library designed to address the low accessibility of Repast for High Performance Computing (RHPC) (81; 82). R4P contains the benefits associated with a high-level programming language, including the wide range of available libraries. The diverse number of libraries enabled the development of a Command Line Interface (CLI) both to streamline the development of the model itself, as well as the various infrastructure needed alongside the simulation, such as experiment generator code, as well as data processing and data visualisation code. Implementing a well-designed CLI, which adhered to best practice principles, significantly streamlined the testing and development process during the development of the ABM (83).

The `SharedContext` object is part of the R4P library. It is where different elements of the simulation can be thought to 'reside'. The `SharedContext` is initialised by the main `Model` class, and during model initialisation, different entities such as the `Board`, `FCT_Communicator`, `FCT_Agent`, and `FundamentalCauseTheory` are placed. It can be thought of as the abstract space in which agents interact with other agents, the environment, their personal 'theory', and the events that are communicated to them.

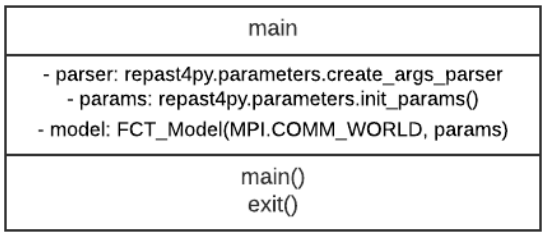


Figure 3.7: *FCT Main Class*

Figure 3.7 shows the `main` class, where the model is initialised, and which calls methods inside the `FCT_Model`. Model parameters are inputted via a `parameters.yaml` file, which contains information such as relevant file locations, output locations, as well as model-based parameters, such as the size of the board, simulation stop time (i.e. k_f) and the parameters outlined in 4.1. Each simulation differs by the set of parameters defined in 4.1.

Message Passing Interface (MPI) is a library used for parallel computation, an approach to running multiple processes simultaneously (84; 85). The FCT ABM doesn't leverage the full use of MPI and parallel computing as the comprehensive implementation of a multi-core approach is outside the scope of the project and not strictly necessary. Thus the FCT ABM

uses a single core during simulation, relying on process management and resource allocation of the operating system of the computer on which the simulation is run; in this case, MacOS (86).

3.2.2 FCT Model

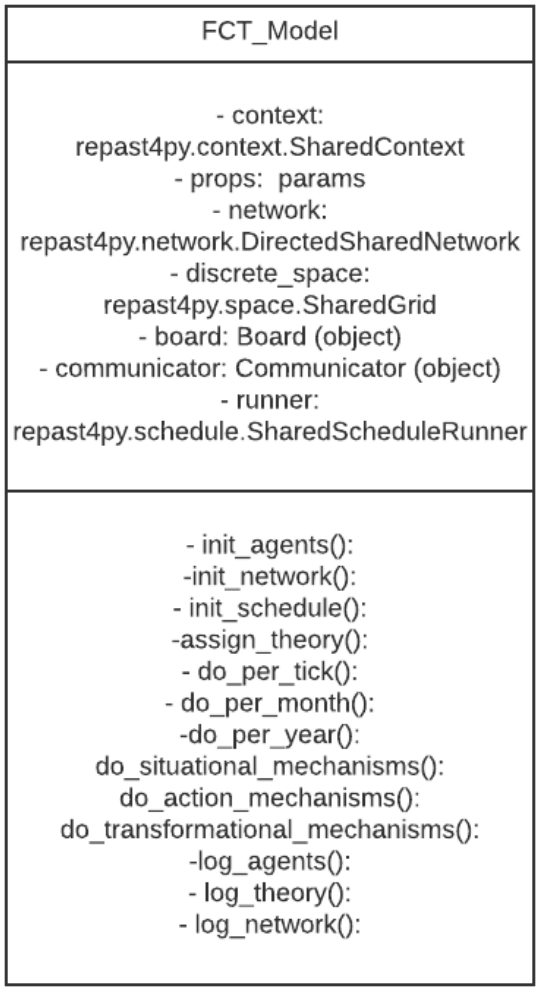


Figure 3.8: *FCT Model Class*

The FCT_Model class is where the simulation is managed. Agents a_i, the board B, and the graph G are all initialised and managed through the `FCT_Model` class. The three key mechanisms of the Coleman Boat, situational, action, and transformational mechanisms, are also managed in FCT_Model and are implemented as the methods `do_situational_mechanisms()`, `do_action_mechanisms()`, and `do_transformational_mechanisms()` respectively. The three Coleman Boat functions contain within them function calls to the `FCT_Agent` and `FCT_Mediator` classes. More specifically, `FCT_Model` manages the interaction between agents and the

`FCT_Communicator` class. `FCT_Communicator`, `FCT_Agent`, and `FCT_Mediator` are discussed in Sections 3.2.3 and 3.2.4.

R4PObjects

The R4P objects are custom classes that are part of the R4P library. These classes include both `repast4py.context.SharedContext`,`repast4py.network.DirectedSharedNetwork`, and `repast4py.space.SharedGrid`, herein referred to as shared context, network, and shared grid, respectively. Shared contexts and projections are fundamental to the workings of R4P.

Both the shared grid and shared network objects are considered as projections in the ABM (81). Agent projections manage the interrelationships between agents (81). Overall, agents can be considered to 'exist' in the context. The network projection is how agents relate to each other, and the shared grid is the environment in which agents live.

A shared context is a custom object which enables developers to place agents in - agents are located inside the context (81). This is the set of all agents $\mathcal{A}$. A shared context object is used to manage processes, which is where the MPI library becomes relevant. In the scenario where agent actions span multiple processes, 'ghost' agents are processed in different process ranks (81). This is how the context object processes agent actions across multiple computer cores. However, this capacity was not fully utilised in the ABM.

The shared network object is an agent network where agents are related based on a network initialisation procedure outlined in Section 3.1.2; originally conceived by Boyd (77). This is the graph G. The initialisation procedure creates agent connections based on DQ, age, education, and the mean weekly units of alcohol an agent consumes. The procedure is intended to create $82\%, 25\%, 5\%, 75\%$, and 15% of connections based on age, DQ, sex, education, and mean weekly units, respectively. The network is a static network, meaning the connections made at the start of the simulation remain throughout. Similarly, the network is directed, meaning connections between agents can either be one-way (unidirectional) or two-way (bidirectional). Agents can only form three connections during the initialisation procedure, a decision taken to reduce the computational overhead. Edges or connections in the network are also weighted by a uniform distribution $U_{(0,1)}$, as outlined in Section 3.1.2. The decision to make a directed, weighted network was taken as it enables a deeper level of customizability compared to an undirected, unweighted network. It also allows provides future developers of the model to modify these parameters should it be relevant.

The shared grid object is a second projection which manages agent interrelationships. The shared grid generates a 50×50 2D discrete space, which can be thought of as the environment in which the agents live. This is the board, or matrix, B. Agents are assigned random locations based on their DQ, with the top left, bottom left, centre, top right, and bottom right being the locations for agents in DQ 1 through 5, respectively. This area assignment is visible both in Figures 5.1a and 3.4. The decision to limit the size of the shared grid to 50×50 cells was taken in part to reduce computational overhead but primarily because the tutorial code for implementing MBSSM with R4P, which modelled the Schelling segregation model, was able to demonstrate segregation adequately using a 50×50 grid. It was therefore assumed that the AHP could also be modelled with a discrete space of the same size.

The `init_schedule()` method is where the R4P scheduler is initiated. The scheduler is another R4P object which tracks what happens as the model runs. This object calls other

functions, such as `do_per_tick()`, `do_per_month()`, and `do_per_year()`. In the context of the simulation, each tick, k, is considered one week.

Description of Other Methods

The `init_agents()`, `init_network()`, and `assign_theory()` methods in the `FCT_Model` class are setup methods used to spawn agents into the shared context, grow the directed network, and assign each agent a theory. In this case, the theory assigned to each agent is a class that takes into account FCT-related parameters, where the FCT-related parameters are based on the agents DQ. The FCT class is described in more detail in Section 3.2.4.

The `do_per_tick()`, `do_per_month()`, and `do_per_year()` methods determine the sequence of events that occur in the model. The R4P scheduler calls these three functions every $k \equiv \mod(1)$, $k \equiv \mod(4)$, and $k \equiv \mod(52)$ respectively. Both the `do_situational_mechanisms()` and `do_action_mechanisms()` methods are called every tick, whereas `do_transformational_mechanisms()` is called every year.

The logging methods `log_agents()` and `log_theory()` log the relevant data for agents and their attached theory object in a `.csv` file. The `log_network()` method stores the agent network at the start of the simulation as a `.graphml` file.

3.2.3 Event Generator

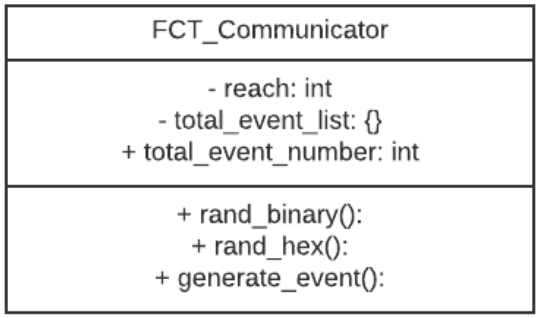


Figure 3.9: *FCT Communicator*

The `FCT_Communicator` is the communicator $\mathcal{C}$, with the `reach` attribute being the maximum size of the subset of communicated agents at any given time step $|\mathcal{A}_{\mathcal{C}}|$. All communicated information is stored in a list `total_event_list`, which is used for monitoring purposes. The attribute `total_event_number` is simply the size of the list of all communicated information. The `FCT_Communicator` class is the macro input to the population of agents, and its communications are the situational mechanisms.

Three methods are present in the `FCT_Communicator` class. The `generate_event()` method manages the information that is outputted; this function returns the event code. The two methods `rand_hex()` and `rand_binary()` generate hexadecimal and binary numbers. Only `rand_binary()` is used in the ABM, but `rand_hex()` is implemented into the `FCT_Communicator` class for extensibility. With further development, hexadecimal informa-

tion could be used to simulate different types of information, possibly harmful information or more valuable information.

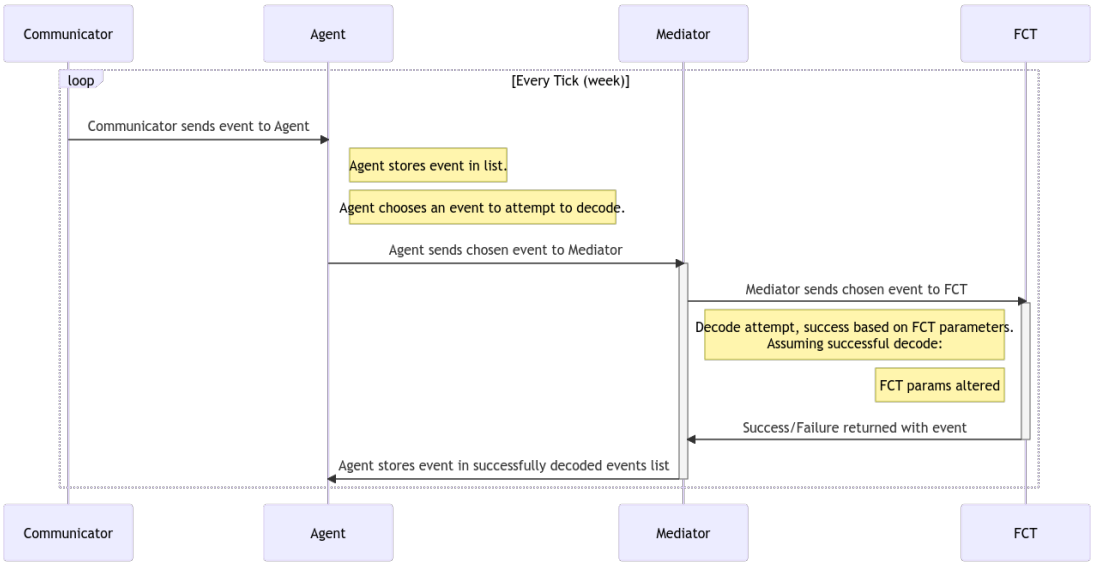
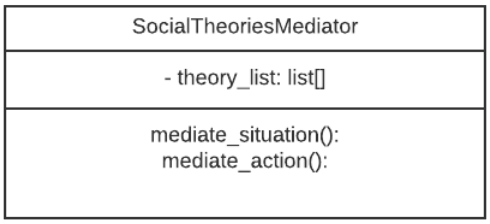


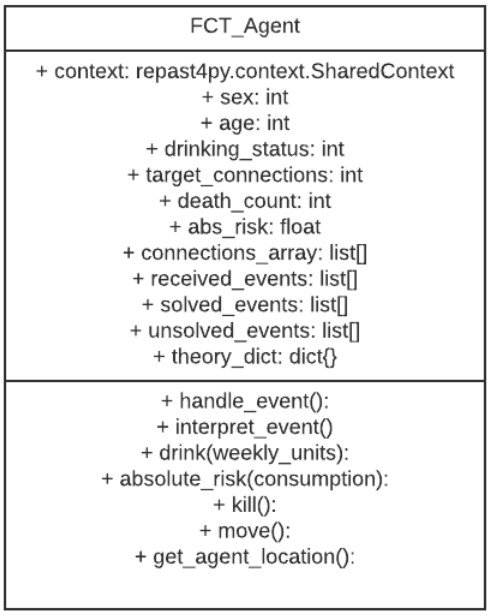
Figure 3.10: *A Sequence Diagram Describing the Interaction Between the Communicator Object and an Agent.*

Figure 3.10 shows the process by which agents receive and process information from the `FCT_Communicator` class. This information is $I(\mathcal{C}(k))$, discussed in Section 3.1.3. The random binary string of length 4 is sent to the agent and converted to a decimal value. The decimal value is then divided by four. The decision to use random binary strings as opposed to decimal values is for extensibility. With further development, the binary-to-decimal conversion process could be used to simulate the agent decoding information in a more realistic sense. Similarly, hexadecimal values could be used to simulate harder-to-decode information. Agents attempt to decode information as described in Section 3.1.4, Equation 3.1. If the sum of an agent's FCT-related resources is greater than the decimal value of the communicated information, the agent can is said to have adapted to that information. Whether an agent adapts to the information or not, the agent must 'pay' an amount of FCT-related resources. However, if an agent is successful in adapting to communicated information, it regains a small amount of knowledge, an FCT-related resource. This process requires the mediator class, a class that manages processes that happen between the agent object and the agent's assigned theory object.

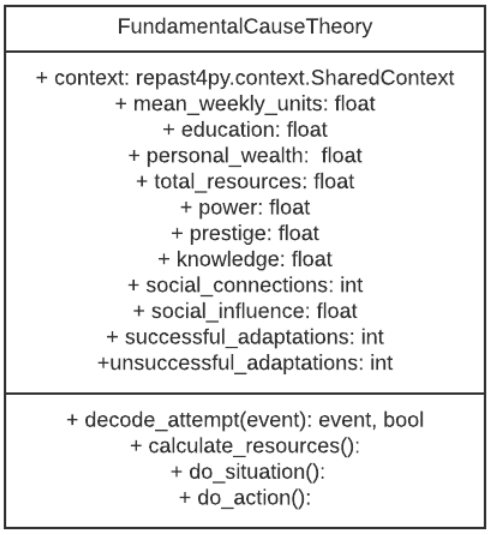
3.2.4 Agents and Theory



(a) *Social Theories Mediator Class*



(b) *FCT Agent Class*



(c) *FCT Class*

Figure 3.11: *Agent, Theory, and Mediator Classes*

Each agent object has a theory object attached via a mediator object. This is a well-known design pattern, and in principle, allows an agent to be attached to a variety of social theories (41; 70). Since the project aims lie in seeing whether an FCT-based ABM would show the AHP, no other social theories were considered. Nonetheless, this explains the `theory_list` attribute in Figure 3.11a.

Figure 3.11b shows the agent attributes and methods. The most notable attributes are `abs_risk` and `drinking_status`. The `abs_risk` attribute is the returned value of the `absolute_risk()` method, which is the risk function outlined in Equation 3.4. The `absolute_risk()` function itself takes the returned value from the `drink()` method, which takes the FCT classes `mean_weekly_units` attribute as an input. The `drinking_status` attribute is an assigned parameter on agent initialisation, where agents are randomly assigned integer values based on a discrete uniform distribution $U_{(0,4;1)}$, where 0 corresponds to no

drinking, and the values $1 \rightarrow 4$ correspond to light, moderate, heavy, and risky drinkers respectively.

The list attributes which keep track of information are `received_events`, `unsolved_events`, and `solved_events`; which correspond to $\mathbb{C}$, $\mathbb{U}$, and $\mathbb{S}$ respectively. The `connections_array` list holds the agents that are connected to a given agent in the static network, and the `target_connections` is an integer that represents the desired number of connections an agent seeks to establish.

Agent methods include `interpret_event()` and `handle_event()`. `interpret_event()` is used to store the information sent by the `FCT_Communicator` object in `received_events`. The `handle_event()` method is used to pass events communicated from other agents into either `sovled_events` or `unsolved_events`, depending on whether the communication was successful or not. Figure 3.12 is the state diagram describing the communication of information $I(a_i)$ between agents. The mathematical description for this process is outlined in Section 3.1.4.

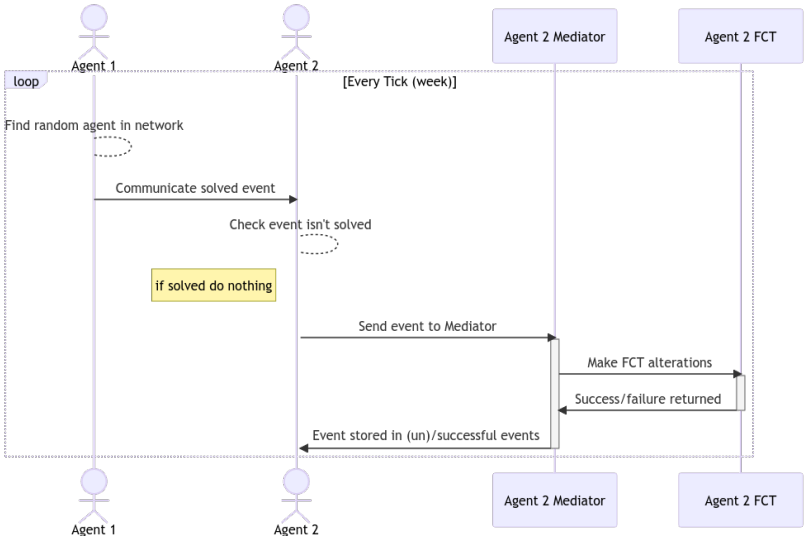


Figure 3.12: *A Sequence Diagram Describing how Agents Communicate Information*

The `FundamentalCauseTheory` class seen in Figure 3.11c contains attributes and methods related to FCT, with attributes such as education, wealth, power, and prestige. The `decode_attempt()` method is where agents attempt to decode and adapt to communicated information $I(\mathcal{C})$. These are the action mechanisms of the simulation. The process which underlies what is demonstrated in Figure 3.12 is found in Equation 3.1, which states that agents need to expend FCT-related resources in order to adapt to communicated information. The mechanisms related to MBSSM, and by extension, the Coleman Boat, are also found in the `FundamentalCauseTheory` class. the methods `do_situation()` and `do_action()` are invoked by the `FCT_Model` class.

3.2.5 Board

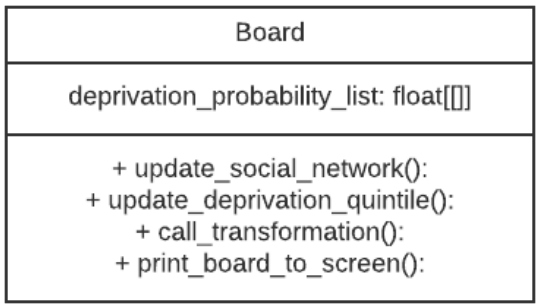


Figure 3.13: *FCT Board Class*

The Board class, seen in Figure 3.13, contains one attribute and four methods. The board class is where the outcomes of the transformational mechanisms are seen. The `deprivation_probability_list` attribute holds the swap probabilities shown in Table 3.2.

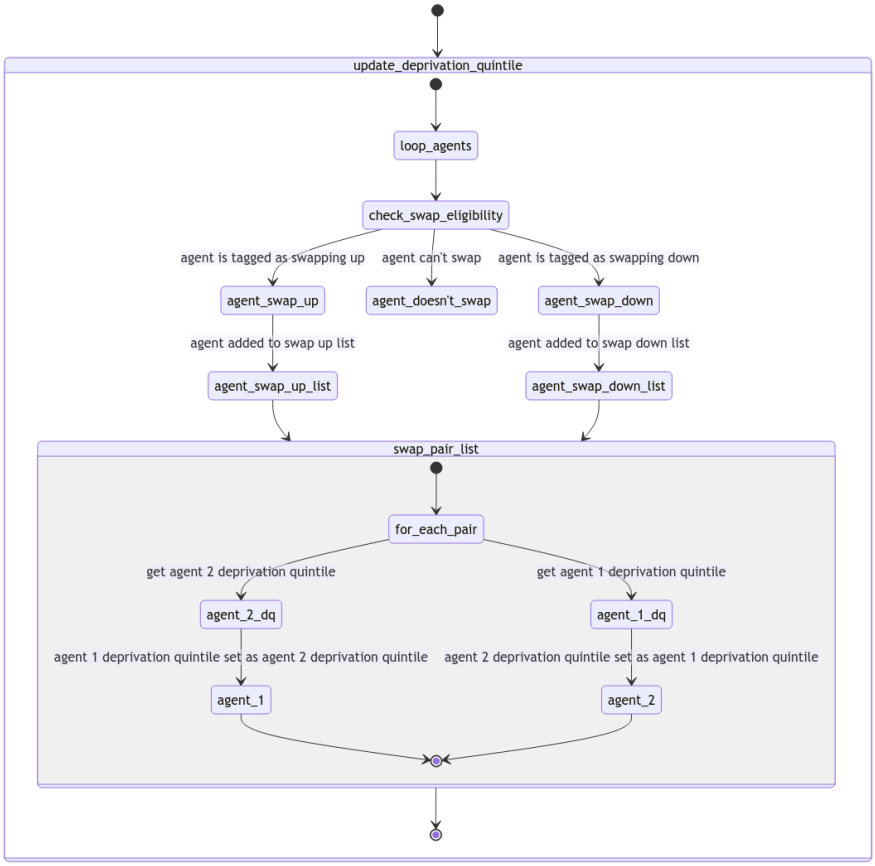


Figure 3.14: *A State Diagram Outlining the Deprivation Swap Process*

Of the four methods, two are noteworthy - `update_deprivation_quintile()` and `call_transformation()`. `call_transformation()` in relation to this model only invokes the `update_deprivation_quintile()` method. However, it is key in the implementation of MBSSM and, by extension, the Coleman Boat. `update_social_network()` was left for extensibility and `print_board_to_screen()` is used for development. Figure 3.14 shows the state diagram describing the deprivation swap process, outlined in Section 3.1.5. In the swap process, every agent in the context is put against the probabilities listed in the deprivation swap table, Table 3.2. Agents are tagged as either swap up or swap down, depending on their current DQ. The agents are then stored in two separate lists, swap up or swap down. If there are more agents that are due to swap up or more agents that are due to swap down, the list is cut such that there are an equal number of agents that swap up and down, satisfying the requirement that $|\mathcal{S}_u| = |\mathcal{S}_d|$ Then, a pair is found for each agent, where the algorithm ensures that there are an equal number of outgoing and incoming agents per DQ.

Chapter 4

Experiments & Methods

4.1 Model Exploration

4.1.1 Latin Hypercube Sampling

In the case of complex models with a high-dimensional parameter space, Latin Hypercube Sampling (LHS) is a valuable technique. It enables modellers to explore various model configurations without necessitating the permutation of all distinct parameters (87). In the case of the ABM outlined in Section 3.2, there is uncertainty around many variables. The full table of input parameters is shown in Table 4.1. Due to the number of variables for which it is uncertain whether they have led to the emergence of the AHP, a LHS is an appropriate technique with which to explore the model in question (87). LHS also requires many fewer simulation runs to complete the full exploration of the parameter space (87). In the context of the FCT model, 100 different samples were generated using the `scipy` Python library. Each sample produces a `.yaml` file, which stores model parameters for a particular simulation. For example, `test_51.yaml` is used for simulation 51. In the context of this project, certain parameters are omitted from the LHS since they bare less relevance when finding the AHP. Overall, the LHS is intended to demonstrate the general behaviour of the model and give insight into expected and unexpected behaviours.

1. Simulation length, $k_f = 540$
2. Grid size $B \in Z^{50\times50}$
3. Random seed number
4. Number of agents $N = 1000$
5. Parameters relating to file locations

4.1.2 Parameter Selection

Φ	Meaning	Parameter Range
ϕ_1	Minimum Age	$\phi_1 \in [16, 21] \in \mathbb{Z}$
ϕ_2	Maximum Age	$\phi_2 \in [82, 115] \in \mathbb{Z}$
ϕ_3	Risk Threshold	$\phi_3 \in [0, 1] \in \mathbb{Z}$
ϕ_4	Successful Adaptation Cost	$\phi_4 \in [0, 1] \in \mathbb{R}$
ϕ_5	Unsuccessful Adaptation Cost	$\phi_5 \in [0, 1] \in \mathbb{R}$
ϕ_6	Successful Adaptation Knowledge Benefit	$\phi_6 \in [0, 0.1] \in \mathbb{R}$
ϕ_7	Communicator Max Reach	$\phi_7 \in [1, 1000] \in \mathbb{Z}$
ϕ_8	β Modifier Male	$\phi_8 \in [0.001, 0.005] \in \mathbb{R}$
ϕ_9	β Modifier Female	$\phi_9 \in [0.0005, 0.0015] \in \mathbb{R}$
ϕ_{10}	β Risk Modifier	$\phi_{10} \in [0, 0.5] \in \mathbb{R}$
ϕ_{11}	Theory Distribution Type	$\phi_{11} \in [1, 3] \in \mathbb{Z}$
ϕ_{12}	Drink Distribution Type	$\phi_{12} \in [1, 2] \in \mathbb{Z}$
ϕ_{13}	Communication Success	$\phi_{13} \in [0, 1] \in \mathbb{R}$

Table 4.1: *Table Outlining Parameters Used in the LHS*

ϕ_1 and ϕ_2

ϕ_1 and ϕ_2 are relevant parameters since they define the range of starting ages at model initialisation. $\phi_1 \in [16, 21]$ since these are common ages for which individuals are legally allowed to purchase alcohol in different countries (88; 89). Similarly, $\phi_2 \in [82, 115]$ relates to the average UK life expectancy and the oldest British person ever to live (90; 91). Agents are allocated age based on this range through a uniform distribution $U_{(\phi_1,\phi_2;1)}$. The overall starting age may have implications on deaths per deprivation quintile since agent risk uses a logistic function which takes into account age (seen in Equation 3.5).

ϕ_3

If the modified risk of an agent, $\bar{\rho}(c, x_1) > \phi_3$, the agent dies. With a lower ϕ_3 value, agents are more likely to die with an increase in alcohol consumption and age. In the model developed by Boyd, this value was set as a random float (77). As such, finding a more concrete value may be of interest.

ϕ_4 and ϕ_5

Adaptation to information costs FCT-related resources, but the exact value of cost for successful or unsuccessful adaptations may vary and should be considered separate. It may be that the model shows the AHP for $\phi_4 \gg \phi_5 \vee \phi_4 \ll \phi_5$, or that it shows the AHP when $\phi_4 \approx \phi_5$.

ϕ_6

Every time an agent successfully adapts to information communicated, whether it be from another agent or the communicator, it gains some knowledge. An agent's total knowledge cannot exceed 1, so the knowledge gained must be small for every successful adaptation, hence $\phi_6 \in [0, 0.1]$.

ϕ_7

The reach of the communicator is the number of agents that can receive information for any given tick. The range of ϕ_7 considers extremities, where there will be some experiments with high quantities of information being sent to the population versus low quantities of information being sent to the whole population. Low quantities of information being sent out would mean agents would have to rely on the spread of information via their social networks. Conversely, high quantities of information being sent out would mean agents will have many opportunities to adapt to information, both from the communicator and their network.

ϕ_8, ϕ_9, and ϕ_{10}

The β modifiers all relate to the two risk functions, $\rho(c)$ and $\bar{\rho}(c, x_1)$. ϕ_8 and ϕ_9 relate to the difference in vulnerabilities to alcohol consumption and are parameterised values based on Scottish data in the simulation performed by Boyd (77). Similarly, ϕ_{10} is a value that exponentiates part of $\bar{\rho}(c, x_1)$ and is a parameterised value in Boyd's simulation. It is, therefore, appropriate to parameterise all three β modifiers.

ϕ_{11} and ϕ_{12}

During model development, it was observed that only utilising uniform distributions for FCT-based parameters would lead to a set of agents whereby the DQ assigned to an agent would only be relevant until the first time the transformational mechanism was performed. Similarly, the range of the distribution for the assignment of mean weekly units was taken based on the model developed by Boyd, which had a cutoff for the maximum mean weekly units of 131.25 (77). As such, grossly unrealistic values were being produced for both FCT-values and mean weekly unit values across the agent population. The Pareto distribution is famously known as being an accurate description of the allocation of wealth in society (92; 93). Similarly, Mulligan states that as societies industrialise and access to cheap technology reaches a large number of people, societies tend toward a curve approximating a normal distribution (known as the Kuznets curve) regarding access to resources (93). As such, $\phi_{11} \in [1, 3]$ means the model will allocate FCT-related resources as a uniform distribution, a normal distribution, or a Pareto distribution, respectively. The Pareto distribution is less relevant to alcohol consumption; as such, $\phi_{12} \in [1, 2]$ switches between a uniform distribution and a normal distribution for the mean weekly alcohol consumption assigned to each agent.

ϕ_{13}

The communication success parameter is related to the communication of information between agents. As outlined in Sections 3.1.4 and 3.2.4; and Equation 3.2, agents have the capacity to communicate solved information with their connections. ϕ_{13} is the requirement that must be met for the agents to successfully adapt to the communicated information. With $\phi_{13} \to 0$, the chance of successful communication between all agents becomes smaller.

4.2 Modification Proposal

It was observed that agent adaptation ratios were being calculated wrong and that the function wasn't resulting in an accurate description of the AHP. Upon further inspection of Equation 3.5, it was observed that the implementation of adaptation ratios may have led to a greater level of modified risk $\bar{\rho}(c, x_1)$ as agents successfully adapted to communicated events, or as $|\mathbb{S}|$ grew. As such, a new equation for calculating adaptation ratios is proposed, Equation 4.3.

The key difference between Equation 3.5 and Equation 4.3 is the change in the ratio regarding the events:

$$(1 - \frac{|\mathbb{U}|}{|\mathbb{C}|} - 0.5)^{\phi_{10}} \to (1 - \frac{|\mathbb{S}|}{|\mathbb{C}|} - 0.5)^{\phi_{10}}$$

It was found that using the ratio us unsuccessful adaptations over total adaptations would lead to a negative modified risk gradient for an increase in unsuccessful adaptations. However, using the ratio of successful adaptations over total adaptations would lead to a positive risk gradient for an increase in unsuccessful adaptations.

Let $|\mathbb{U}|,\ \phi_{10},\ \rho(c),\ x_1$ be some constant positive number. Then

$$\bar{\rho}(c, x_1) = \left(1 - \frac{|\mathbb{U}|}{|\mathbb{C}|} - 0.5\right)^{\phi_{10}} \tag{4.1}$$

$$\text{leads to } \begin{cases} \frac{\partial}{\partial c}\bar{\rho} \leq 0, & \text{for } 0 < |\mathbb{S}| \leq |\mathbb{U}| \\ \frac{\partial}{\partial c}\bar{\rho} > 0, & \text{for } |\mathbb{U}| < |\mathbb{S}| < \infty \end{cases} \tag{4.2}$$

However, the proposed modification is

$$\bar{\rho}(c, x_1) = \left(1 - \frac{|\mathbb{S}|}{|\mathbb{C}|} - 0.5\right)^{\phi_{10}} \tag{4.3}$$

$$\text{leads to } \begin{cases} \frac{\partial}{\partial c}\bar{\rho} \geq 0, & \text{for } 0 < |\mathbb{S}| \leq |\mathbb{U}| \\ \frac{\partial}{\partial c}\bar{\rho} < 0, & \text{for } |\mathbb{U}| < |\mathbb{S}| < \infty \end{cases} \tag{4.4}$$

The difference between Equations 4.1 and 4.3 lies in the way adaptations relate to risk. In Equation 4.1, as agents solve more events, that is, as $|\mathbb{S}|$ grows, the risk gradient of the modified risk function with respect to consumption $\frac{\partial}{\partial c}\bar{\rho}$ grows, which implies that as the agents successfully adapt to public health interventions, they are exposed to more risk. By contrast, Equation 4.3 ensures that as agents successfully adapt to public health interventions, that is, as $|\mathbb{S}|$ grows, the risk gradient $\frac{\partial}{\partial c}\bar{\rho}$ shrinks, implying they are less exposed to alcohol-related harms.

4.3 Searching for the AHP

The experiment setup consists of two separate sets of model runs. The LHS generated `.yaml` files remains the same over both experiment sets. Experiment set 1 runs the model outlined in Section 3.2 using the risk equation outlined in Equation 3.5. Experiment set 2 runs the same model but with the modified risk function proposed in Equation 4.3. To find the AHP, three conditions must be met, a positive death gradient, a negative adaptation gradient, and a negative consumption gradient. As such, new observational variables are defined in Table 4.2.

Variable	Definition	Descripiton
$\vec{\tau_1}$	Death Toll	Vector of observed death counts per DQ across the whole simulation
$\vec{\tau_3}$	Adaptation Ratio	Vector of adaptation ratios per DQ according to Equation 4.1 or 4.3
$\vec{\tau_3}$	Consumption	Vector of observed consumption per DQ across the whole simulation
Δ	Deprivation Quintile	A single column matrix representing DQs
k	Time Steps	A discrete variable representing time steps
γ_i	Experiment Gradient	A gradient obtained through a least squares approximation

Table 4.2: *Key Observational Variables Used in Data Analysis*

Each experiment gradient is calculated using a least squares approximation, given in Equation 4.5. The experiment gradient γ_i refers to the gradients obtained relating to the death toll, adaptation ratio, and consumption for γ_1, γ_2, and γ_3 respectively.

$$\gamma_i \coloneqq \beta_1 \iff \min ||\vec{\tau_i} - \Delta\vec{\beta}||^2 \tag{4.5}$$

4.3.1 Positive Death Gradient

First, there must be a positive death gradient with respect to the deprivation quintile over the whole simulation. This indicates that with an increase in deprivation, an agent is more likely to be exposed to alcohol-related harms, the key feature of the AHP (6; 94). Hence, the condition $\gamma_1 > 0$ must be met.

4.3.2 Negative Adaptation Gradient

Second, there must be a negative adaptation gradient. This implies that with an increase in deprivation, agents struggle to adapt to communicated information $I(\cdot)$ due to lack of FCT-related resources. The negative adaptation gradient also points toward the correct implementation of FCT within the model. Hence, the condition $\gamma_2 < 0$ must be met.

4.3.3 Negative Consumption Gradient

Finally, there must be a negative consumption gradient. The presence of a negative consumption gradient is a key characteristic of the AHP (6; 94). The negative consumption gradient implies that with an increase in deprivation, there is a decrease in the amount of alcohol consumed on average, per week. Hence, the condition $\gamma_3 < 0$ must be met.

Chapter 5

Results

The results section contains an overview of the simulations, first showing figures that are relevant to all simulations. Three simulations are shown from the two experiment sets, one set of results from experiment set 1, and two sets of results from experiment set two. The results are shown in the following sections of this chapter are the simulations for which γ_1, γ_2 and γ_3 most closely resembled the AHP.

The process to run each simulation within each experiment set was automated, where a CLI command was used to run the model through the different `.yaml` files created by the LHS. The same `.yaml` files were used for both experiment sets. After gathering each of the outputs from the individual simulations, the algorithm seen in Listing 5.1 would then calculate γ_1, γ_2, and γ_3 for all experiments within the experiment set, and return an ordered list dependent on γ_1 and γ_2.

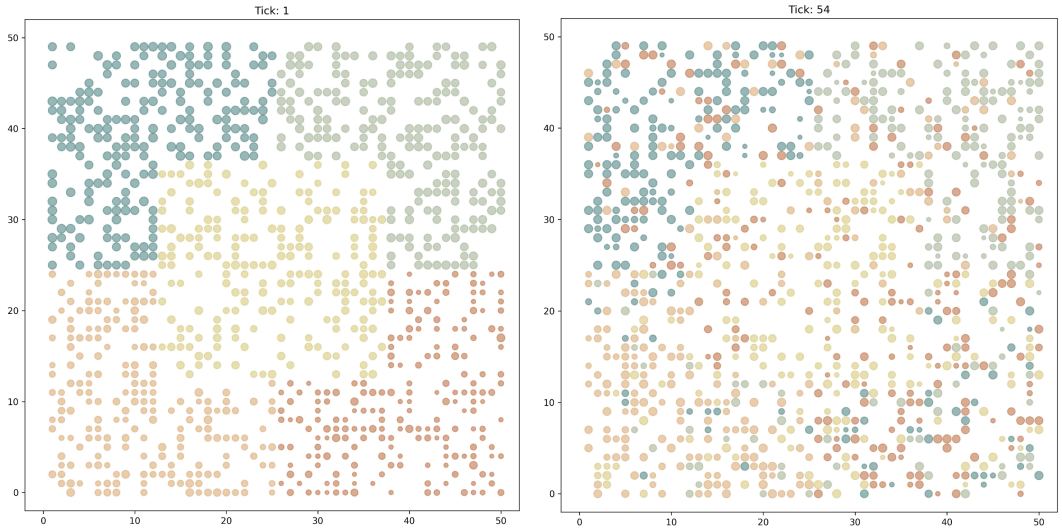
Listing 5.1: Sorting gradients

```
sorted_gradients = sorted(gradients, key=lambda x: (x[1], x[2]), reverse=True)
```

Visually inspecting 100 separate simulations per experiment set is an inefficient process to find the AHP, hence the automation of the process. The ordered gradient table was then visually inspected to find candidate experiments that demonstrated the AHP. A separate Dash app was built to visualise the data to give additional context for each of the candidate simulations (95). The full table of gradients for both experiment sets is found in Appendix C and D for experiment sets 1 and 2 respectively. The experiments shown in the main body of the report are highlighted in blue in tables C.1 and D.3. The full dataset was stored in a MySQL database using an automated process which took the `.csv` outputs of each simulation and passed them into the database. A MySQL database is a convenient way to quickly scan through and sort data using database queries (96). Using the Sequel Ace app, simulations were searched to check for reasonable values of variables, for example, checking for `NaN` values (97).

5.1 Simulation Visualisations

5.1.1 Board



(a) *Agent Starting Position on the Board* **(b)** *Agents on the Board, Swapping at Tick 54*

Figure 5.1: *Agents on the Board*

5.1.2 Agent Network Topology

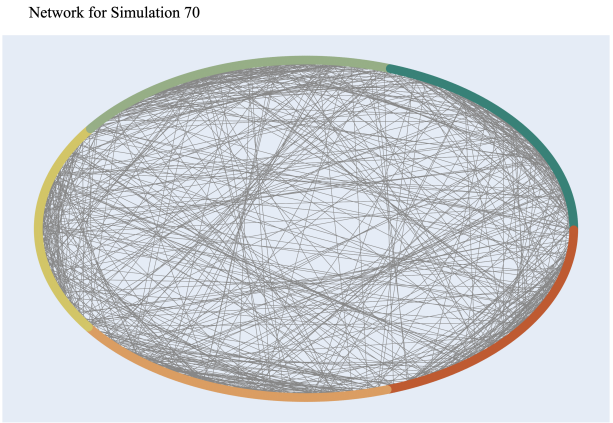


Figure 5.2: *Starting Network Topology for Every Simulation*

Statistic	Value
Vertices $\lvert\mathbb{V}\rvert$	1000
Edges $\lvert\mathbb{E}\rvert$	2998
Density $\frac{\lVert G\rVert}{\binom{\lvert G\rvert}{2}}$	0.003
Diameter $diam(G)$	14
Average Clustering	0.011
Average Degree $\epsilon(G)$	5.996
Average Katz Centrality	0.0316

Figure 5.3: *Network Statistics*

5.1.3 Commentary

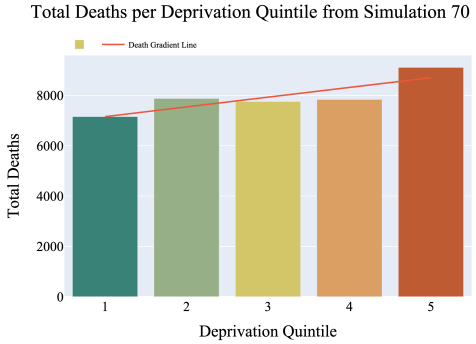
Board

Using the same colour scheme outlined in Figure 3.4, Figure 5.1 shows agents on the discrete space, with colour representing DQ, and agent size representing the total FCT-related resources available to the agent. On visual inspection of figure 5.1a, it is seen that agents in DQ 1 have the largest amount of FCT-related resources, whilst agents in DQ 5 have the least. This can be interpreted as visual validation of FCT being properly implemented into the model. Similarly, Figure 5.1b shows the tick in which agents swap, where agents lie in their previous DQ before moving to their newly assigned DQ. Here, it can be seen that agents are adhering to the swap probabilities outlined in Table 3.2, where agents from DQ 3 have a higher probability of swapping down $P_s(a) = 0.4$ than agents from DQ 1, hence why there are more agents are painted red in the centre quintile than in the top left quintile, where DQ 1 is. Similarly, Agents from DQ 5 have the highest probability of swapping up $P_s(a) = 0.6$ hence why Figure 5.1b shows a relatively low number of agents in the bottom right quintile painted red after the swap pairs have been allocated.

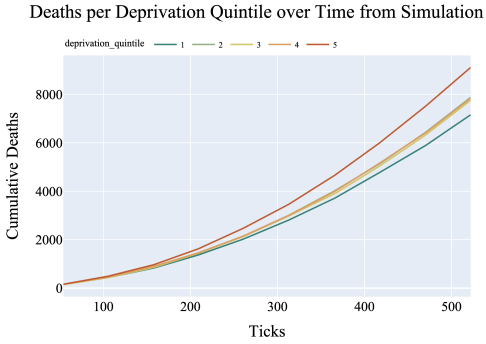
Network

Each simulation follows the same network initialisation protocol outlined in Section 3.2.2. The initial network setup is the same for every simulation run. Figure 5.2 shows the starting graph topology for every simulation, and Figure 5.3 shows some of the extremal properties associated with the initial graph. The density of a graph is the proportion of potential edges over the total number of edges (98). It is not surprising that the requirement for each agent to only have 3 connections, when there is a population of 1000 agents, leads to a sparse graph with an overall density of 0.003. The diameter of the graph is 14, meaning the greatest distance between any two vertices or agents is 14 edges (98). A graph density value of 0.003 isn't surprising given the total number of edges as 1000 and the requirement for 3 connections per agent, outlined in Table 3.1. Visually inspecting the graph on Figure 5.2, it can be seen that there is a high level of inter-DQ connectivity, where agents of the same DQ have formed connections with other agents in the same DQ. Similarly, there is a high level of intra-DQ connectivity, indicating agents have formed connections based on similarities in attributes such as age, sex, education, and mean weekly units of alcohol consumed. The inter-intra-DQ connectivity can be considered as a validation of the proper implementation of the network setup procedure, outlined formally in Section 3.1.2 and outlined in software in Section 3.2.2.

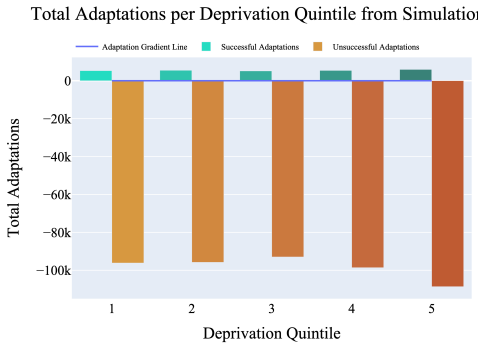
5.2 Experiment Set 1: Original Risk Function



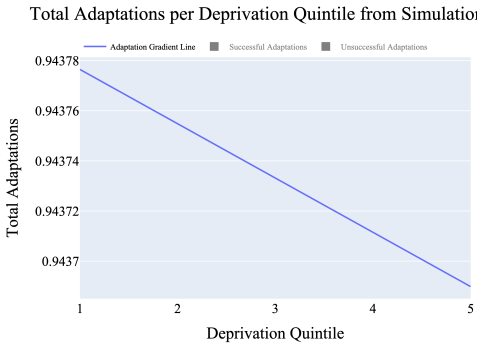
(a) *Total Deaths by DQ time, Experiment Set 1, Simulation 70*



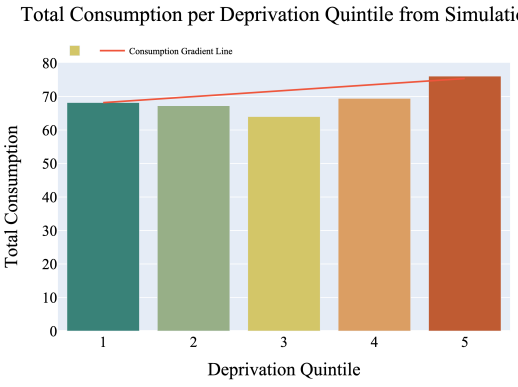
(b) *Deaths over time in Experiment Set 1, Simulation 70*



(c) *Successful & Unsuccessful Adaptations per DQ in Experiment Set 1, Simulation 70*



(d) *Adaptation Gradient in Experiment Set 1, Simulation 70*



(e) *Total Consumption and Gradient by DQ in Experiment Set 1, Simulation 70*

Figure 5.4: *Full set of Results from Experiment Set 1, Simulation 70*

5.2.1 Commentary

γ_i	Result
γ_1	389.6
γ_2	-2.16×10^{-5}
γ_3	1.79

Table 5.1: *Experiment Set 1 Simulation 70 γ values*

With the 100 simulations performed, simulation 70 from experiment set 1 showed the most reasonable set of data. While many other experiments showed $\gamma_1 > 0$ and $\gamma_2 < 0$, results that may imply the presence of the AHP, when observing the figures corresponding to those simulations, it was clear that while many simulations demonstrated the AHP with appropriate outcomes between DQ 1 and DQ 5, the quantity of deaths, adaptations, and consumption in DQs $2-4$ didn't accurately reflect the AHP. This was usually due to a situation where, for example, DQ 4 distinctly had many more deaths than DQ 5 or 3.

Deaths

Figures 5.4a and 5.4b show that the simulation is accurately representing the differential harm outcomes of high versus low DQ agents. What made death data from simulation 70 compelling is the relatively similar total deaths for DQs 2, 3, and 4, all the while there being differential outcomes between DQ 1 and 5. The value for γ_1 is relatively high as a result.

Figure 5.4b shows how deaths change with time. Deaths and ticks seem to increase proportionally together, suggesting some kind of power law relationship. Simulation 70 was one among many simulations to demonstrate this relationship, and interestingly simulations with resulting gradients close to the AHP also demonstrated this relationship.

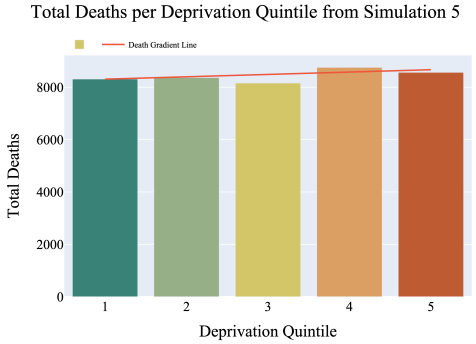
Adaptation

Figures 5.4c and 5.4d show the agents adapting to information, with differential outcomes between DQ 1 and DQ 5. The resulting γ_2 value is negative, however, the actual value is so small it is almost negligible and could, in effect, be considered a straight line.

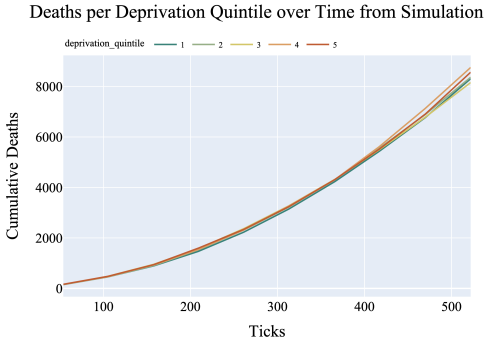
Consumption

Figure 5.4e shows the differences in consumption between the different DQ groups. Here there is a significant positive gradient, which is contrary to the AHP. Also Figure 5.4e shows a significant difference in consumption between DQ 3 and DQ 5.

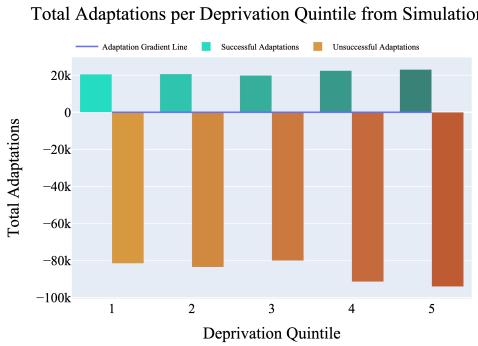
5.3 Experiment Set 2: Modified Risk Function



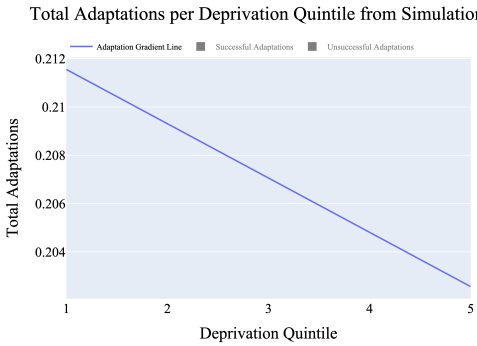
(a) *Total Deaths by DQ time, Experiment Set 2, Simulation 5*



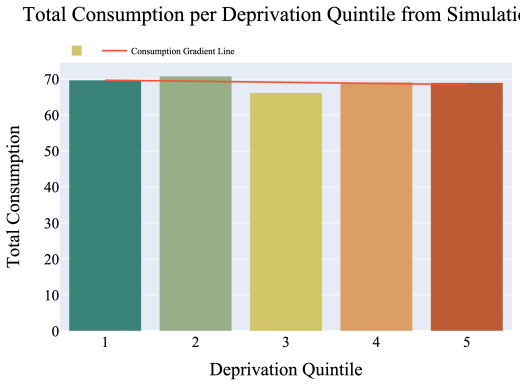
(b) *Deaths over time in Experiment Set 2, Simulation 5*



(c) *Successful & Unsuccessful Adaptations per DQ in Experiment Set 2, Simulation 5*



(d) *Adaptation Gradient in Experiment Set 2, Simulation 5*



(e) *Total Consumption and Gradient by DQin Experiment Set 2, Simulation 5*

Figure 5.5: *Full set of Results from Experiment Set 2, Simulation 5*

5.3.1 Commentary

γ_i	Result
γ_1	90
γ_2	-0.225
γ_3	-0.303

Table 5.2: *Experiment Set 2 Simulation 5 γ values*

Simulation 5 from Experiment Set 2 satisfies the requirements to demonstrate the AHP. Simulation 5 showed γ values that appeared to show the AHP, seen in Table 5.3. As the results seemed to demonstrate the AHP, the decision was taken to re-run simulation 5 over an extended time period, setting $k_f = 1200$. The results of this extended simulation 5 are shown in Section 5.4.

Deaths

Figure 5.5a shows $\gamma > 0$, satisfying the first requirements for finding the AHP. However, on further inspection, it is clear that the total death toll for agents in DQ 5 is less than the total death toll for agents in DQ 4. This in itself doesn't reflect the AHP, both in terms of expected outcomes and in reality, as is shown in Figure 1.1. Figure 5.5b shows a similar death over time trend as Figure 5.4b, where deaths seem to grow in proportion with time.

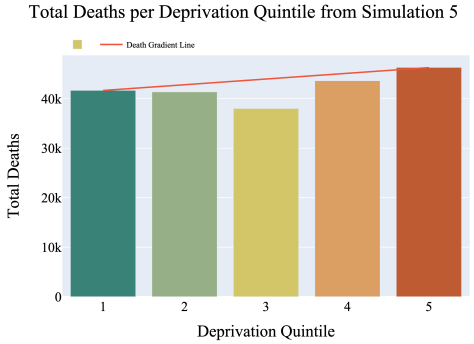
Adaptation

Figures 5.5c and 5.5d show an appropriate, negative adaptation ratio $\gamma_2 < 0$ satisfying the second requirement for the AHP. The γ_2 value found in Table 5.3 is four orders of magnitude larger than the γ_2 value found in simulation 70, experiment set 1. This suggests that the change in the adaptation ratio calculation may have been effective in giving more reasonable outputs relating to adaptation.

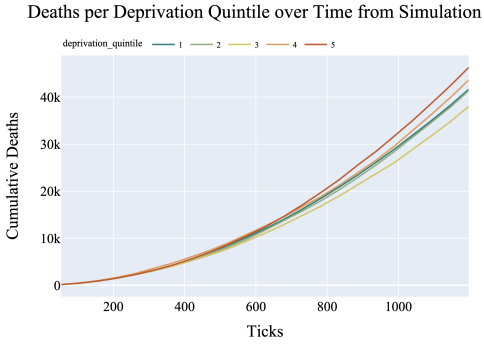
Consumption

Figure 5.5e shows a negative consumption gradient. Although simulation 74 from experiment set 2 also demonstrated a negative consumption gradient, there were significant differences in the overall consumption levels across DQs. It could be argued that the dip in consumption for DQ 3 is enough to discount the results, but regardless, upon examination of the entire dataset, the observed γ_2 values did not appear to deviate excessively from expected norms. The data presented a reasonable negative gradient that was just as plausible as a potential positive gradient.

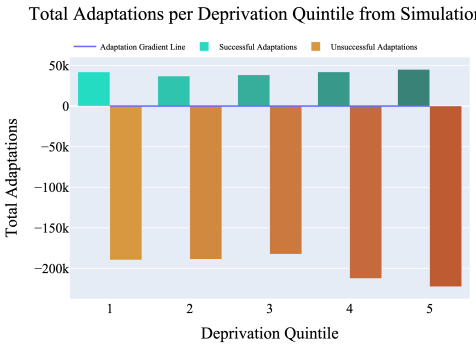
5.4 Extending Simulation 5 from Experiment Set 2



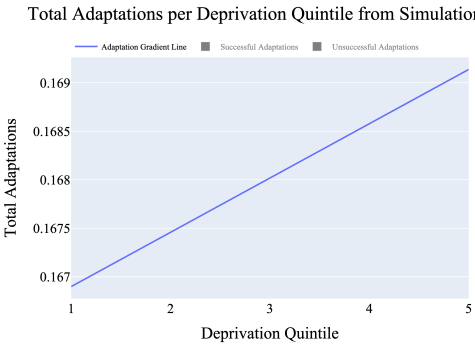
(a) *Total Deaths by DQ time, Experiment Set 2, Extended Simulation 5*



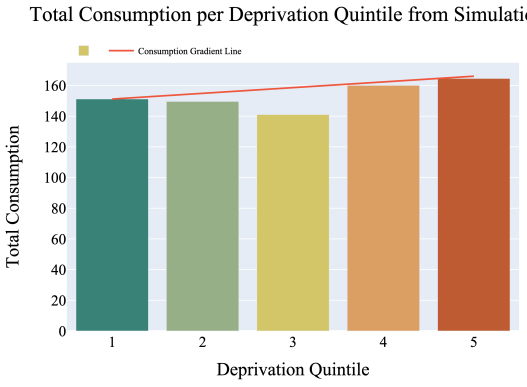
(b) *Deaths over time in Experiment Set 2, Extended Simulation 5*



(c) *Successful & Unsuccessful Adaptations per DQ in Experiment Set 2, Extended Simulation 5*



(d) *Adaptation Gradient in Experiment Set 2, Extended Simulation 5*



(e) *Total Consumption and Gradient by DQ in Experiment Set 2, Extended Simulation 5*

Figure 5.6: *Full set of Results from Experiment Set 2, Extended Simulation 5*

5.4.1 Commentary

γ_i	Result
γ_1	170.7
γ_2	0.0019
γ_3	1.7933

Table 5.3: *Experiment Set 2 Extended Simulation 5 γ values*

The expectation with extending simulation 5 from 520 ticks to 1200 was that the gradients associated with the AHP would become more pronounced. Extending simulation 5 shows that the variable most significantly impacted by time is death. Since the AHP is demonstrable across scales and has been demonstrated across large amounts of time, it could be considered to be time-invariant. It was expected that with a larger k_f other AHP-related results would become more pronounced; however, the values for γ_i in extended simulation 5 suggested the AHP is time invariant in its implementation in the FCT ABM. Overall, despite using the same `.yaml` file, hence the same parameter settings, the extended version of simulation 5 wasn't able to successfully demonstrate the AHP.

Deaths

Figure 5.6a shows deaths per DQ and 5.6b shows deaths over time k. These results show that the extended timeline was able to demonstrate a more pronounced impact of alcohol consumption on agent deaths. It also shows that death trends are not strictly time-varying, and there is some level of consistency in differential outcomes with respect to deaths. γ_1 for extended simulation 5 is greater than γ_1 for original simulation 5, suggesting that differential death outcomes remain consistent across time for the same parameter setup.

Adaptation

Figure 5.6c shows agent adaptations by DQ, and Figure 5.6d shows a positive adaptation gradient. A positive γ_2 for the extended simulation indicates the AHP isn't present.

Consumption

Figure 5.6e shows total consumption by DQ. The consumption gradient γ_3 is also positive, further indicating that the extended version of simulation 5 didn't show the AHP. The range of γ_3 remains consistent with other consumption gradients found through both experiment sets.

Chapter 6

Discussion

6.1 Differential DQ Outcomes

6.1.1 Showing the AHP through FCT

Significant time and effort was spent diagnosing issues regarding the emergence of differential outcomes between DQs. Experiment sets 1 and 2 were the last sets of simulations to run and generated the most reasonable datasets. Many experiments in experiment sets used during development didn't demonstrate the AHP; only simulation 5 from experiment set 2 showed gradients γ_i where a case could be made for the AHP being shown. Nonetheless, the resulting differential harm outcomes that make the AHP such a striking phenomenon in reality, were not seen in simulation 5, as values for γ_2 and γ_3, although negative and positive respectively they were not significantly high. The intention behind extending simulation 5 was to see if the AHP would become more pronounced with more time. That is, with $k_f = 1200$, the AHP should have become more apparent, the assumption being γ_i vales have some time scaling attribute. This reasoning was in part due to the potential power law relationship between cumulative deaths per DQ ($\vec{\tau}_1$ and time k. Since simulation 5 showed cumulative deaths of agents in DQ 5 growing faster than for agents in other DQs, it was assumed that with more time steps, this would become more pronounced alongside the other gradients γ_2 and γ_3. Although this proved to be correct, with a larger γ_1 for extended simulation 5, this was not true for γ_2 and γ_3.

Φ_5	Φ_{70}	Meaning
$\phi_1 = 20$	$\phi_1 = 19$	Minimum Age
$\phi_2 = 112$	$\phi_2 = 102$	Maximum Age
$\phi_3 = 0.064128$	$\phi_3 = 0.20809$	Risk Threshold
$\phi_4 = 0.720235$	$\phi_4 = 0.642612$	Successful Adaptation Cost
$\phi_5 = 0.669016$	$\phi_5 = 0.47031$	Unsuccessful Adaptation Cost
$\phi_6 = 0.035906$	$\phi_6 = 0.025479$	Successful Adaptation Knowledge Benefit
$\phi_7 = 664$	$\phi_7 = 91$	Communicator Max Reach
$\phi_8 = 0.002011$	$\phi_8 = 0.001061$	β Modifier Male
$\phi_9 = 0.000834$	$\phi_9 = 0.001061$	β Modifier Female
$\phi_{10} = 0.394013$	$\phi_{10} = 0.417671$	β Risk Modifier
$\phi_{11} = 3$	$\phi_{11} = 1$	Theory Distribution Type
$\phi_{12} = 1$	$\phi_{12} = 1$	Drink Distribution Type
$\phi_{13} = 0.120978$	$\phi_{13} = 0.099509$	Communication Success

Table 6.1: *Table Outlining Parameters Used in Simulation 5 and Simulation 70*

Although all parameters for Φ_5 and Φ_{70} are at least somewhat relevant, the focus lies in parameters with large relative differences between simulation 5 and simulation 70.

ϕ_3

ϕ_{3_5} and $\phi_{3_{70}}$ differ by one order of magnitude. The risk threshold parameter relates to how much risk an agent can 'tolerate' before it dies.

ϕ_4 and ϕ_5

ϕ_{4_5} and $\phi_{4_{70}}$ are within the same order of magnitude, as are ϕ_{5_5} and $\phi_{5_{70}}$. Nonetheless, in both cases, the successful adaptation cost is higher than the unsuccessful adaptation cost. The difference between ϕ_4 and ϕ_5 is greater in simulation 70 than for simulation 5, which may be contributing to the difference in γ_2 values, $\gamma_{2_{70}} \ll \gamma_{2_5}$.

ϕ_6

ϕ_6 for both simulations is of a relatively similar value. Hence, in both simulations, agents had a relatively similar benefit upon the successful adaptation to communicated information, with simulation 70 having a slightly lower benefit.

ϕ_7

ϕ_{7_5} and $\phi_{7_{70}}$ differ by one order of magnitude. As established in section 6.1.2, information spread evenly and quickly throughout the network and has a high output from the communicator, as in simulation 5, it could be considered that agents were 'flooded' by information relative to agents in simulation 70, giving lots of opportunity to adapt both successfully and unsuccessfully.

ϕ_8, ϕ_9, and ϕ_{10}

ϕ_{10_5} and $\phi_{10_{70}}$ differ by one order of magnitude. ϕ_8 and ϕ_9 lie in the same order of magnitude for both simulations. Hence, the difference in harms experienced by male and female agents differs significantly in simulation 5 when compared to simulation 70.

ϕ_{11} and ϕ_{12}

Simulation 5 uses a Pareto distribution in the allocation of FCT-based parameters, whereas simulation 70 uses a uniform distribution. Both simulations assign mean weekly units of alcohol consumed by the population as uniform distributions.

ϕ_{13}

Inter-agent communication in simulation 5 has $\approx 12\%$ chance of being successful, whereas, in simulation 70, it's $\approx 1\%$.

6.1.2 Implications of Φ Values

Simulation 5 and simulation 70 can be thought of as societies in two separate parallel realities. In simulation 5, agents live in a risky world where adaptations are expensive, but they gain more knowledge when adapting to information. Information, however, is relatively easy to communicate and spreads quickly and in large volumes throughout society. Male and female agents experience significantly different risks associated with alcohol consumption. Finally, in simulation 5, there is an uneven distribution of resources.

In simulation 70, agents still live in a risky world, but agents are more tolerant to risk than in simulation 5. It costs less to adapt to information in simulation 70 than in simulation 5, and it costs more to adapt successfully than unsuccessfully and less knowledge is gained upon a successful adaptation. Information still spreads relatively fast once it enters society, but information enters society in smaller volumes. Information is harder to communicate than in simulation 5. Male and female agents experience the same risk from alcohol, and resources are distributed relatively evenly across society. The quantity of alcohol consumed by the average agent in both societies is the same.

Given these two separate realities, it comes as no surprise that simulation 5 would see an adaptation gradient many orders of magnitude higher than simulation 70. The higher difference in cost for successful adaptations versus unsuccessful adaptations in simulation 70 may be attributed to the much lower success rate for adaptations, seen in Figure 5.4c, and consequently, explains the greater number of successful adaptations in simulation 5. The greater number of successful adaptations in simulation 5 could also be attributed to the volume of information being passed to agents from the communicator. Information is also more easily comprehended in simulation 5, contributing to the greater number of successful adaptations.

Since alcohol consumption is effectively the same in both simulations, the difference in consumption gradients could be attributed to a combination of the swapping method employed and how the populations are assigned their alcohol consumption levels. Since agents only choose to swap up or down, and alcohol consumption is uniformly distributed, stochas-

ticity in the swapping algorithm could be attributed to agents who consume more being part of DQ 1 and agents who consume less being part of DQ 5.

Although more information is spreading faster and in larger volumes in simulation 5, agents in lower DQs have far fewer resources than in simulation 70 due to the Pareto distribution used to allocate resources. This, and the relatively strong negative adaptation gradient, can be attributed to the agent death gradient present in simulation 5. The death gradient present in simulation 70 is stronger, and this could be attributed to the larger difference between the cost of successful versus unsuccessful adaptations. For further work, the AHP may become more apparent from implementing the improvements suggested in Section 6.2, taking Φ_5 values as a starting point, and running an optimisation procedure.

6.1.3 Extending Simulation 5

Complex adaptive systems are non-linear and time-varying (99). This nonlinearity leads to emergent behaviour that isn't necessarily predictable. It was expected that with an extension of simulation 5, the AHP would become more pronounced. The death gradient did become more pronounced, however, the result of extended simulation 5 was that the AHP overall became less pronounced. This could be attributed to two characteristics of the model, the swapping mechanisms, and the nature of the network.

Swapping Mechanisms

Swapping mechanisms don't account for the current DQ of an agent, nor their available FCT-related resources. It is based on probability. As such, agents with zero wealth, for example, may still swap with an agent with 1 wealth from DQ 5 into DQ 1. This is analysed further in Section 6.2.

High Intra-DQ Connectivity

High intra-DQ connectivity effectively means that agents in a given DQ communicate with agents within other DQs frequently. This means that when an agent from a high DQ decodes information and attempts to pass it on to other agents, it is more likely than not to pass that information onto agents outside its own DQ. Therefore, the spread of information is not kept within DQs, and it has a tendency to spread across the network quickly. This is analysed further in Section 6.2

6.1.4 Verifying the Power Law, Death-Time Relationship

Power laws are characterised by proportional changes in two related variables. A power relationship appeared seemingly consistently with simulations that were close to demonstrating the AHP. The simulations shown in Figures 5.4, 5.5, and 5.6 all demonstrated what seemed like a power law relationship between k and $\vec{\tau}_1$. Power laws can be verified by plotting the variables on a log log scale and checking for a linear relationship (100).

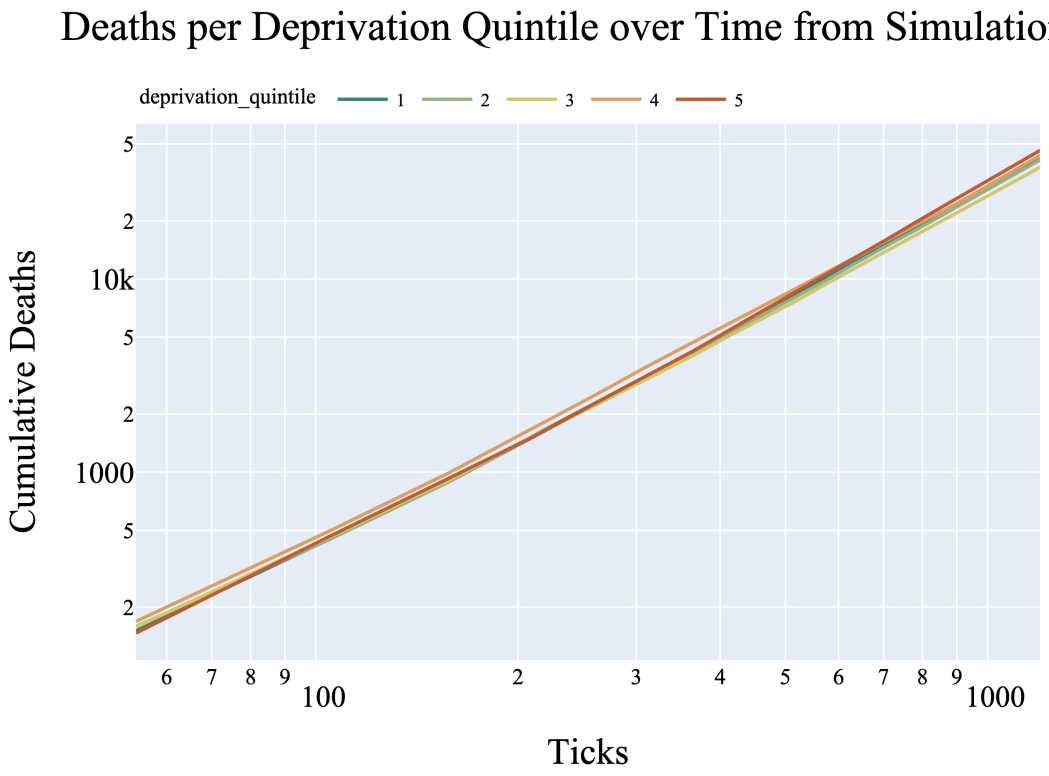


Figure 6.1: *A* $\log - \log$ *Scale Applied to Figure 5.6b*

Since agents don't exit the simulation when they die and only their age resets, the agent death rate is inherently unrealistic. In fact, looking at Figure 6.1, the entire agent population would have died within the first $k = 100$ in extended simulation 5 if agent death had been modelled more realistically. Power laws may occur due to feedback loops, but also multiplicative processes and fractal processes (101). What gives rise to the power law may not be of high relevance, but Figure 6.1 does demonstrate a higher gradient value for DQ 5 than for the other DQs which is a characteristic that could be used for developers searching for the AHP. Power laws are scale-invariant (101). Testing for scale invariance is beyond the scope of the dissertation but may be relevant for further development.

6.2 The MBSSM Implementation

Although results from two sets of experiments are shown, 5 different experiment sets were generated overall. The initial three experiment sets were used for the development of the various search algorithms and the development of the interactions between the outputted data and the MySQL database. These initial three experiment sets were used to ensure the proper software implementation of the situation, action, and transformational mechanisms outlined in Sections 3 and 4. The third preliminary set of experiments was where the issue with the adaptation ratio, outlined in Section 4.2, was observed through the sorting of data via Sequel Ace. The majority of adaptation gradients for experiment set 1 showed adaptation gradients in the $\times 10^{-3}$ and $\times 10^{-4}$ range, which, when compared to the actual values of `solved_event` and `unsolved_events` didn't make sense. Agents with a high `death_count` had a low value `solved_events`, which was contradictory. The risk function was plotted using the Desmos graphing calculator, verifying the issue outlined in Equation 4.1 and led to the proposal seen in Equation 4.3 (102).

6.2.1 Situational Mechanisms

The situational mechanisms are seen to be working adequately in the model, with agents having information communicated to them via the `FCT_Communicator`. Information of different decimal values is being adapted to differently and appropriately by agents in different DQs, shown in the number of negative adaptation gradients $\gamma_2 < 0$ present in both experiment sets. This confirms the intended emergence, where agents from DQ 1 adapt to information more frequently than agents from DQ 5. This verifies that the interaction between the `FCT_Communicator` and the `FCT_Agent` is working as intended.

Negative Information

Alcohol advertising has been found to reduce avoidance bias toward alcohol consumption in heavy drinkers, as well as contribute toward excessive underage alcohol consumption (103; 104). Alcohol advertising is arguably more visible in Western societies compared to alcohol-related public health campaigns. However, more money was invested in public health than in alcohol advertising in 2020 (105; 106). The alcohol industry spent £285 million in 2020 in advertising, and the UK government spent £3.2 billion in the same year; some fraction of which will be allocated toward alcohol-related public health campaigns (105; 106). Since it is established that there exist situational mechanisms which expose an individual to more alcohol-related risk, information which contributes to increasing agent risk could be implemented in the ABM. Let $|\mathcal{R}|$ be the sum of risky information an agent receives. let $\mathcal{R}, \mathbb{U}, \mathbb{S} \in \mathbb{C}$. The risk function may then be modified further to include $|\mathcal{R}|$ as follows:

$$\bar{\rho}(c, x_1) = \rho(c) \cdot (1 - \frac{|\mathbb{S}| - |\mathcal{R}|}{|\mathbb{C}|} - 0.5)^{\phi_{10}} \cdot \frac{1}{1 + e^{-\frac{x_1 - 45}{9.5}}} \tag{6.1}$$

6.2.2 Action Mechanisms

Action mechanisms involve the communication of information between agents and agents attempting to adapt to information communicated from the `FCT_Communicator`. Agents were verified to be communicating successfully through testing during development. Results from Figures 5.4, 5.5, and 5.6 show that agents generally failed to adapt more than they adapted; in general, results show $|\mathbb{U}| \gg |\mathbb{S}|$. This was somewhat intentional since information can be hard to synthesise in reality, but the large disparity between $|\mathbb{U}|$ and $|\mathbb{S}|$ was not. Although $|\mathbb{U}| > |\mathbb{S}|$ should be true in general, the disparity indicates an issue with the implementation. This may be attributed to the low communication success value of $\phi_{13} = 0.120978$ for simulation 5 experiment set 2, and $\phi_{13} = 0.099509$. for simulation 70, experiment set 1. However, the disparity could also be attributed to the stochasticity in the randomly assigned binary number. Likely the cause can be attributed to a combination of the two. The negative γ_2 values for simulation 70, experiment set 1 and simulation 5, experiment set 2 are indicators that the AHP can be displayed through FCT in the ABM. However, the positive γ_2 for extended simulation 5 further indicates stochasticity in the the model.

Network Implementation

The ABM developed by Boyd used a dynamic network which reinitialised at each $k \equiv \mod 52$ (77). The network used in the ABM presented in this dissertation is static. There are significant differences between static and dynamic networks and relevant implications in the context of an ABM built with FCT. The network initialisation procedure, outlined in Sections 3.1.2 and 3.2.3, generate agent connections based on age, DQ, sex, education and mean weekly units. Since agents swap DQs as part of the swapping mechanisms outlined in Sections 3.1.5 and 3.2.5, a static network corresponds to agents keeping connections with agents which lie outside of their own DQ. Agents that swap from a low DQ to a high DQ don't form connections with agents in the higher DQ; hence they don't gain the benefits of agents which are more likely to decode information due to the higher `total_resources` available to the agents in the higher DQ. Similarly, agents moving to a lower DQ don't experience the cost of being linked with agents who have less `total_resources` available to them. The implications of the static network are such that agents are, effectively, stuck with the social circle they started with, which keeps agents who started in DQ 1 in a beneficial position relative to the population, and consequently, agents in DQ 5 in a detrimental position relative to the population.

Nonetheless, the capacity to visualise the results of the network setup procedure, seen in Figure 5.2, and to gain statistics related to the procedure, seen in Figure 5.3, gives interesting insights. The diameter of the network, $diam(G) = 14$, can be thought of as the furthest possible distance between any two agents in the network. This means that from the point some information $I(\mathcal{C}(k))$ is sent from the communicator to an agent, the minimum possible timesteps k it would require for that information to be received by the whole network (given that the communication success rate $\phi_{13} = 1$) is $k = 14$. Considering $I(\mathcal{C}(k))$ is sent from the communicator to a subset of agents $|\mathcal{A}| = \phi_7 \in [1, 1000]$ at every timestep k, even for $\phi_7 = 1$, for a high communication success rate $\phi_{13} \simeq 1$, information spreads quickly across the network. Since the average degree $\epsilon(G) \simeq 6$, nearly all agents have three incoming and three outgoing connections, meaning information spreads not only quickly but also evenly

through the network.

Different network topologies, including topologies which consider disparate centrality between agents, could yield interesting insights. The centrality of an agent in a network can be thought of as its importance (107). The Katz centrality measures the importance of an agent based on the importance of the agents it is connected to (107; 108). The average Katz centrality seen in Figure 5.3 was the same for all agents, at 0.0316. Hence every agent in the network was equally as important as any other agent. Network topologies such as scale-free and hierarchical networks show differential outcomes for individual agent centrality measures; few agents have many important connections, and many agents have few important connections (109; 110). These network topologies could be used to implement a more differentiated network structure; Highly influential agents in DQ 1 may have influence not only agents within DQ 1 but also agents in DQs $2 \rightarrow 5$. This would model something akin to an 'influencer'. Similarly, agents in DQ 5 may have a high centrality within DQ 5 but a low centrality relative to the rest of the network, modelling people who are influential in their community but not across the whole society.

Changes to Education, Wealth, and Consumption

Education typically doesn't remain constant over an individual's lifetime. Although individuals who start life in a more deprived do experience less attainment in school and university relative to their less deprived peers, mature students often pursue higher education because of their own aspirations and have the capacity to do so thanks to access courses (111–113). In the ABM presented, an agent's education stayed constant throughout, even after death. Education and the related knowledge variable, which is an FCT-based parameter, relevant in the risk calculation.

As with education, individual wealth doesn't remain static over their lifetime. On a population level, an individual's wealth tends to increase with age (114). This is not accounted for in the ABM, where wealth remains relatively constant. Wealth being an FCT-based parameter also contributes to the risk calculation via the capacity to adapt to communicated information.

Finally, alcohol consumption and the impact alcohol has on the body change with age. Older individuals drink fewer units per week than younger individuals but drink more frequently (115; 116). Older individuals are impacted more by alcohol consumption than younger individuals, hence a higher exposure to risk (116). This isn't accounted for in the ABM, since as with other FCT-related parameters, mean weekly units of alcohol remain static. However, the risk function outlined in Equation 3.5 does account for age through the logistic function.

Education, wealth, and mean weekly units could be implemented dynamically, where these parameters could be made to change with time. As an approximation, education, wealth, and mean weekly units could be implemented through a logistic function as follows:

$$f(x) = \frac{L}{1 + e^{-k(x - x_0)}}$$

where x is age, x_0 is the average of the possible age ranges of the simulation $\frac{x_{youngest} + x_{oldest}}{2}$. k is some arbitrary growth rate and would likely be a parameterised value. L is the maximum possible value of $f(x)$ and could be determined by using DQ $L = \frac{DQ}{5}$. This would, however,

limit agents in lower DQs significantly, but in principle, DQ should be used to represent some sort of limitation in this case. $f(x)$ could then be used to represent how wealth and education change with age, and $-f(x) + L$ be used to represent mean weekly units changes with age. This is by no means a concrete method to accurately represent real-world dynamics, but it may contribute to increasing model accuracy.

6.2.3 Transformational Mechanisms

The board shown in Figure 5.1b demonstrates the DQ swapping mechanisms are being performed as intended. Similarly, the visually inspecting Figure 5.1a demonstrates that the allocation of FCT-based resources in the agent initialisation algorithm outlined in Section 3.2.2 works as intended, with agents in DQ 1 being larger (hence having more FCT-based resources) and agents from DQ 5 being smaller (hence having less FCT-based resources). Having a small γ_2 implies there is little difference between how agents in DQ 1 and DQ 5 adapt to communicated information, both from agents in their network and from the communicator. γ_2 being a small value may be attributed to how agents swap using the method outlined in Section 3.1.5, using Table 3.2.

Deprivation Swap Model

The deprivation swap model used by Boyd, outlined in Table 3.2, Section 3.1.5, is based on a model proposed by Georgi et al. (77; 80). Georgi states that the model itself was based on the assumption that people move DQs based solely on education and starting families (80). In reality, this isn't strictly true. A 2020 report by the Social Mobility Commission found that individuals from low DQ areas moved for education, work prospects, to maintain social connections, and quality of life improvements (117). As such, there are numerous reasons why an individual from any DQ may move to any other DQ, likely not limited to the reasons outlined in the Social Mobility Commission's report.

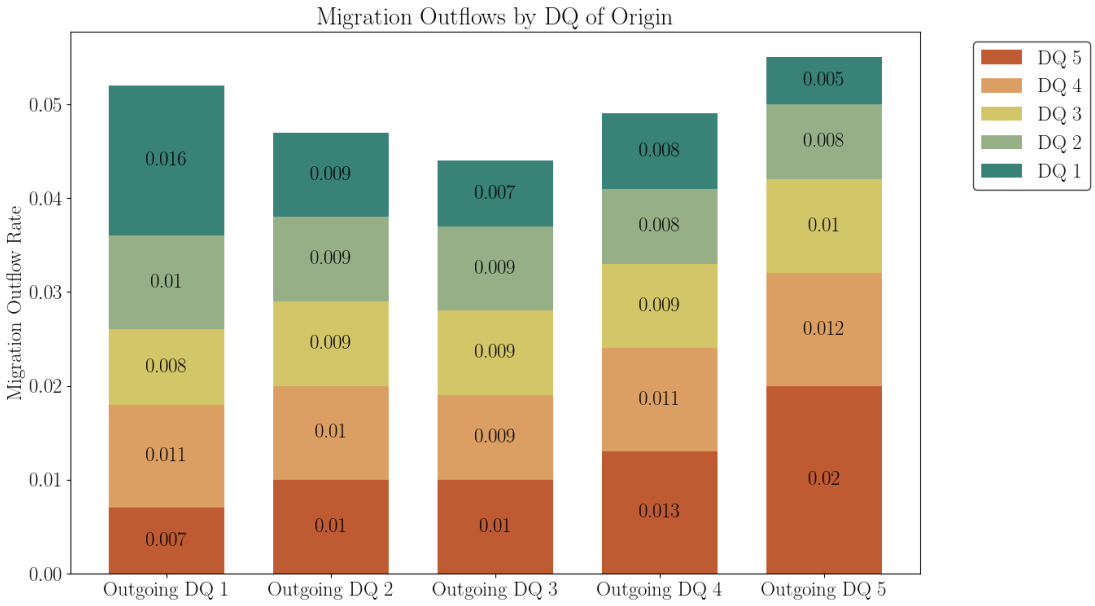


Figure 6.2: *Migration Outflows by Area of Origin* [1]

An agent's DQ is central to both the AHP and FCT, hence a large determining factor in their exposure to alcohol-related harms. Agents using the method outlined by Boyd swap deprivation quintiles adequately, although the ABM only considers whether agents swap up or down DQ. A more realistic approach would consider agents moving to a specific DQ, based on their current DQ. Figure 6.2 could be used to implement such a model, where the current DQ of an agent could be used to determine which DQ they are more likely to move to. It seems reasonable that agents from DQ 1 are more likely to stay in DQ 1; similarly, agents from DQ 5 are more likely to stay in DQ 5. Although Figure 6.2 shows migration flow rates, these could be normalised into swap probabilities and implemented in a modified swap table. As such, for the ABM presented in this dissertation, the implementation of a separate deprivation swap table, based on real-world deprivation swap probabilities between different DQs may lead to increased reliability of the model reproducing the AHP.

[1] Adapted from Social Mobility Commission, 2020 (117)

Chapter 7

Conclusion

Alcohol is one of the world's most commonly consumed drugs and arguably one of the most harmful. The AHP shows that the harmful effects of alcohol disproportionately impact those most vulnerable, those low in SES. FCT, as a theory, has the potential to explain the AHP due to its emphasis on how SES relates to the risk of exposure to disease. Previous work by Boyd placed FCT into a mechanistic context, allowing for the development of an ABM. The previously ABM developed by Boyd was abstracted and replicated, placing the original model based on Scottish census data into a generalised context. MBSSM was used as a software architecture to facilitate placing the social mechanisms present in FCT into an ABM, modelling situational, action, and transformation mechanisms, representing public health campaigns, agents attempting to adapt to those health campaigns, and the changes the campaigns lead to in the context of the whole population. During development, an issue was identified with the original implementation of the risk function, where agents experienced less risk with less successful adaptations. A new approach to the risk function was proposed, whereby agents who successfully adapted experienced less risk from alcohol consumption.

100 samples of the 13-dimension parameter space were generated via a LHS. Two experiment sets were then run, the first set using the old approach to the risk function and the second set using the altered risk function, both with the same LHS samples. Results from each set were stored, and an algorithm searched through each simulation to find death, adaptation, and consumption gradients that would indicate the AHP. Overall, one experiment demonstrated the correct set of gradients, and accompanying data, for which it could be said that the simulation had shown the AHP, simulation 5 from experiment set 2. When running simulation 5 over a longer time span, however, the AHP was not shown, suggesting issues with the implementation of the ABM. Since the AHP is not a time-varying phenomenon, extending the timeline of the simulation should yield more pronounced results.

To yield more consistent results, model alterations suggested included adding more realism to the situational, action, and transformation mechanisms. Modelling alcohol advertisements as detrimental to agents may lead to further disparity in risk outcomes. Making the agent network represent a hierarchical structure may lead to disparity in how information is spread around the population. Finally, adapting the swap method to replicate real-world migration flows may yield a more consistent emergence of the AHP. In all, the ABM developed presents a good starting point, but more work is needed if the model is to consistently reflect the complexity and nuances surrounding the AHP.

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Appendices

Appendix A

Gantt Chart Amendments

Appendix B

MBSSM Architecture

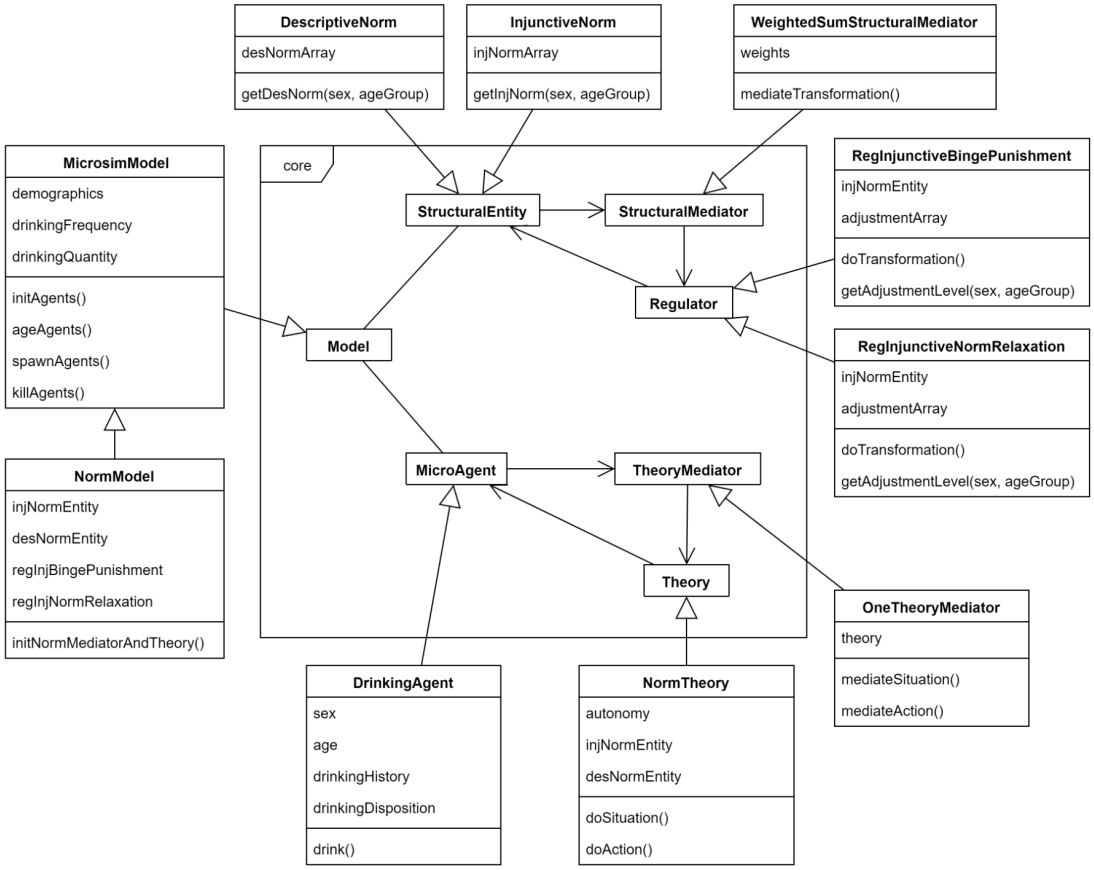


Figure B.1: *MBSSM Applied to a Social Norms Model* [1]

[1] Original by Vu et al. (41)

Appendix C

Experiment Set 1: Gradient Results

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
42	471.7	-0.00289038	2.67928409
69	406.7	-0.00073692	1.38447558
79	401.0	-0.00088578	2.88016327
70	386.9	-2.16629599e-05	1.79615919
51	376.3	-0.00081015	1.05921874
95	374.4	-0.00031857	2.61194983
57	354.1	-3.78657374e-05	2.54800691
6	339.6	-0.00203447	2.91275525
52	332.5	-0.00102016	2.09664718
16	323.1	0.0001026	3.62610129
76	321.6	-0.00067751	0.88403153
82	276.8	-0.00071115	1.17164243
29	257.1	-0.00195546	1.86658295
27	253.9	-0.00150234	0.49610453
44	253.0	0.00025558	0.65803284
38	250.6	-0.00143186	2.19116949
17	247.0	-0.00059605	1.00256715
85	244.2	-0.0002876	1.55476807
93	242.5	-0.00108275	0.79601988
99	235.8	-0.00110101	2.27318322
84	235.2	-0.00066224	1.60014581
72	233.5	0.0002737	0.5879153
65	228.2	-0.00035489	0.55435014
74	225.6	0.00039949	2.01422176
47	220.1	-0.00110093	1.73090088
3	220.0	9.01009571e-05	0.65918158
7	219.7	-0.00229672	1.54154261
77	214.9	-0.00100431	1.96675775
63	212.8	-0.0005435	0.91327309

Table C.1: *Experiment Set 1 Gradient Results, Part 1*

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
10	209.2	-0.00106914	1.62377041
97	207.8	-0.00014341	0.58793853
32	206.5	-0.00011996	0.4157514
20	204.1	-0.00136855	1.3037022
86	203.7	-0.00081691	2.12881355
80	203.7	-0.00239732	1.71712004
14	192.8	-7.39460333e-05	0.84977327
81	192.0	-0.00014472	1.03277327
41	191.5	-0.00124507	1.38120995
48	188.4	-0.0003729	0.42327702
56	187.6	-0.00145019	1.41786934
78	182.9	0.00060574	0.68991286
91	180.7	-0.00026313	2.2612603
25	178.9	0.00024899	1.17619444
31	178.5	0.00011924	0.70584516
62	177.6	-0.00097257	0.62279783
89	176.7	-0.0010204	1.28834379
39	175.4	-0.00015598	1.38933218
36	171.0	-0.00176086	0.32559396
40	170.9	-0.00274693	1.3645346
92	170.3	-0.00134559	1.20247278
46	169.7	-0.00041519	0.83217556
23	169.0	-0.00020831	0.45467755
88	164.8	-0.00015122	0.84132199
49	164.1	-0.00041072	0.27923875
61	162.9	-0.00042206	1.15759331
9	159.8	-0.00064021	0.36873743
5	158.8	-0.00046348	0.96032777
64	157.1	-0.00185787	0.29693237
11	156.6	-0.00025967	0.56115786
4	155.4	-9.28848957e-05	0.62807956
94	151.5	-0.00174043	1.04741976
67	149.5	-0.00165964	0.36601391
98	148.2	-0.00264905	0.52240789
13	147.9	2.65281554e-05	1.30838899

Table C.2: *Experiment Set 1 Gradient Results, Part 2*

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
2	142.8	-0.00074772	0.73465844
87	142.4	-0.00056874	0.8688565
60	139.6	-0.00291384	0.33529703
33	138.3	3.74815956e-05	0.19520434
24	135.4	-0.00053285	0.95752892
71	134.1	-0.0006658	1.32439023
83	130.9	-0.00242079	0.90233644
21	128.2	-4.45449302e-05	0.68851861
96	123.5	-0.00229409	0.67603194
68	122.3	-0.00040751	0.88349933
18	120.6	-0.00020507	0.51455497
100	108.5	-0.00052594	0.41522045
90	105.7	-0.00109438	0.94136907
15	102.9	-0.00096355	0.5797905
30	102.5	-0.00179304	0.24523551
53	101.1	-3.56676414e-05	1.30906951
26	99.8	-0.00019049	0.40620639
1	95.6	-0.00142579	1.07306283
45	91.4	-0.00015605	0.11361168
58	89.2	-0.00011398	0.53502079
55	88.8	-0.0010209	0.55985696
43	83.3	-0.00081035	0.13421907
66	73.8	-0.00115121	0.39312099
54	70.8	-0.00203322	0.04297549
22	70.0	-0.0001577	0.47519175
12	67.7	-0.00236552	-0.50829488
28	65.2	-0.00030549	0.41844048
8	63.6	-0.00035853	0.79736089
34	51.7	-0.00036977	0.37220835
37	49.6	-0.00111368	0.34346116
35	46.9	-0.00227693	0.71692097
73	36.8	-0.00020245	0.15532128
50	27.9	-0.00064728	0.00029772
75	21.9	-0.00063599	0.58892398
19	19.2	-0.00058379	0.05608921
59	16.0	-0.00089191	0.37516115

Table C.3: *Experiment Set 1 Gradient Results, Part 3*

Appendix D

Experiment Set 2: Gradient Results

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
17	454.7	3.64680824e-05	1.18658305
86	443.2	-0.00202891	1.8213978
78	425.2	-0.0008357	3.01817328
62	405.4	-0.00169849	0.90819921
99	385.9	2.02253112e-05	3.31791061
85	382.1	-0.00027513	2.26645855
93	302.0	-0.00399094	0.77385918
36	294.3	-0.00445756	0.52377871
84	290.6	-0.00063786	3.39339003
73	286.7	-0.00022305	1.20261152
28	279.9	-0.0004677	1.05073354
95	277.9	-0.0004271	2.43454037
71	274.9	-0.00065582	2.25575909
79	269.7	-0.0007158	1.99478225
60	261.3	-0.0006517	0.70622667
42	256.7	-0.0014309	1.66950675
27	254.5	-0.00092543	0.6426757
97	251.0	-0.00040384	0.48176758
44	245.9	-0.00112509	1.01810231
92	234.8	-0.00062401	2.00397783
1	233.6	-0.00117019	1.55655781
53	229.5	-0.00027559	1.29735553
18	225.8	-0.00018096	0.71339384
56	224.3	-0.00068312	2.26515368
72	223.9	-0.00069166	0.28304089
80	222.1	-0.00194557	1.8368404
81	221.1	-0.00069103	1.40957276
57	219.9	-0.00064324	1.43028389
82	219.2	-0.00044908	1.3424276
13	219.1	-0.00094978	1.33417061
16	217.4	-0.00073966	1.82345098
41	216.1	-0.00017779	1.92716667

Table D.1: *Experiment Set 2 Gradient Results, Part 1*

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
47	213.8	1.08281	0.77910653
34	209.4	-0.00201104	0.82902658
67	208.9	-0.0013483	0.74737267
40	207.4	-0.00180175	1.57323222
19	202.6	-0.00029681	0.78556292
65	202.0	-0.00038451	0.81473108
94	200.8	-0.00179796	1.50311085
70	199.8	-8.08103838e-05	0.70590754
76	194.7	3.42483564e-05	0.6382319
3	191.3	-0.00039445	0.65940658
21	189.7	-0.00045133	0.66078697
55	189.6	-0.00033111	1.76093259
14	189.5	-0.00097128	0.4026903
2	189.3	-0.00095759	1.46097693
10	188.7	-0.0013329	1.82590336
77	179.9	-0.00058151	0.69748155
59	172.0	-8.68879908e-05	0.97860252
6	170.7	-0.00191232	1.79337131
46	170.4	-0.00013823	1.6389807
32	168.4	-0.00109831	0.54811138
49	165.5	-0.00017459	0.10473547
64	161.4	-0.00062392	0.42598796
22	154.9	-0.00069038	0.81116953
51	154.4	-0.00149715	0.46565954
24	149.0	-0.00051468	1.51201936
83	147.2	-0.0020246	0.55355843
90	147.0	-8.85716283e-05	0.85724612
52	146.1	-0.00145532	0.62742733
39	145.2	-0.00166839	1.09135897
89	143.4	-0.00130603	1.22093288
87	141.6	-0.00056258	0.8347024
20	141.3	-0.00176128	0.32568354
4	140.0	-0.00031398	1.20519171
45	129.0	-0.00016175	0.48375265
68	126.8	-0.00028079	0.76759761
15	126.4	-0.0012899	0.41160729

Table D.2: *Experiment Set 2 Gradient Results, Part 2*

Simulation	Death Gradient	Adaptation Gradient	Consumption Gradient
69	120.2	-0.00019775	0.38064034
26	114.7	-0.00014693	0.75658629
38	113.9	-0.0025645	0.42220982
63	110.0	-0.0003023	0.50043154
29	109.0	-0.00052855	0.77272246
7	98.5	-0.00066093	1.32540354
35	96.6	-0.00286018	0.7903826
54	93.7	-0.00377592	0.32114809
5	90.0	-0.00224724	-0.30315585
43	86.3	-0.0006416	0.37475419
66	85.8	-0.00167605	0.77557601
50	85.5	-0.00088593	0.21620459
74	83.2	-0.00291823	-0.12617468
91	82.0	-0.00017713	1.4305857
30	82.0	-8.76759002e-05	0.61526895
48	78.1	-0.0010542	0.19178651
37	75.2	-0.00074597	0.54802517
100	67.3	-0.00038134	0.70118611
12	65.6	-0.00031077	0.39648042
11	60.2	-5.00798375e-05	0.33983878
98	59.6	-0.00163424	0.53996107
8	44.4	-0.00021789	0.51363474
58	42.1	1.11836972e-05	0.23590365
75	40.4	-0.00060689	0.53082225
9	24.9	-0.00053355	0.52652041
33	17.1	-0.00155447	0.07525847
25	10.1	-1.13320841e-05	0.3841704
88	9.0	-0.00020197	0.73143864
61	-18.8	-0.0001706	1.81946141
31	-28.5	-0.00094716	0.18256497
23	-52.5	-0.00112442	0.32146983
96	-73.1	-0.00176647	-0.1269245
5 **longer**	170.7	0.00191232	1.79337131

Table D.3: *Experiment Set 2 Gradient Results, Part 3*

Appendix E

Simulation 5 YAML File

```
agent.logger: FCT_Model/outputs/agent_logger_out.csv
agent.props.file: FCT_Model/props/agents/agent_props.json
area.file: FCT_Model/props/area/DQ_areas.csv
beta.modifier.female: 0.000834
beta.modifier.male: 0.002011
board.props.file: FCT_Model/props/area/DQ_areas.csv
board.size: 50
communication.success: 0.120978
communicator.max.reach: 664
communicator.min.reach: 50
correlation.parameter: 1
count.of.agents: 1000
drink.distribution.type: 1
knowledge.gain: 0.882835
max.age: 112
min.age: 20
network.file: FCT_Model/props/network/graph.txt
network.file.updated: FCT_Model/props/network/graph_updated.txt
random.seed: 1
resource.depletion: 0.842568
risk.modifier.beta: 0.394013
risk.threshold: 0.064128
stop.at: 520
successful.adaptation.cost: 0.720235
successful.adaptation.knowlege.benefit: 0.035906
theory.distribution.type: 3
theory.logger: FCT_Model/outputs/theory_logger_out.csv
theory.props.file: FCT_Model/props/agents/theory_props.json
threshold: 0.5
unsuccessful.adaptation.cost: 0.669016
```

Appendix F

Simulation 70 YAML File

```
agent.logger: FCT_Model/outputs/agent_logger_out.csv
agent.props.file: FCT_Model/props/agents/agent_props.json
area.file: FCT_Model/props/area/DQ_areas.csv
beta.modifier.female: 0.001318
beta.modifier.male: 0.001061
board.props.file: FCT_Model/props/area/DQ_areas.csv
board.size: 50
communication.success: 0.099509
communicator.max.reach: 91
communicator.min.reach: 50
correlation.parameter: 1
count.of.agents: 1000
drink.distribution.type: 1
knowledge.gain: 0.812645
max.age: 102
min.age: 19
network.file: FCT_Model/props/network/graph.txt
network.file.updated: FCT_Model/props/network/graph_updated.txt
random.seed: 1
resource.depletion: 0.337623
risk.modifier.beta: 0.417671
risk.threshold: 0.20809
stop.at: 520
successful.adaptation.cost: 0.642612
successful.adaptation.knowlege.benefit: 0.025479
theory.distribution.type: 1
theory.logger: FCT_Model/outputs/theory_logger_out.csv
theory.props.file: FCT_Model/props/agents/theory_props.json
threshold: 0.5
unsuccessful.adaptation.cost: 0.47031
```

Appendix G

GitHub Repository Link

Github link, click: here.